

ILLINOIS TECH

Armour College of Engineering

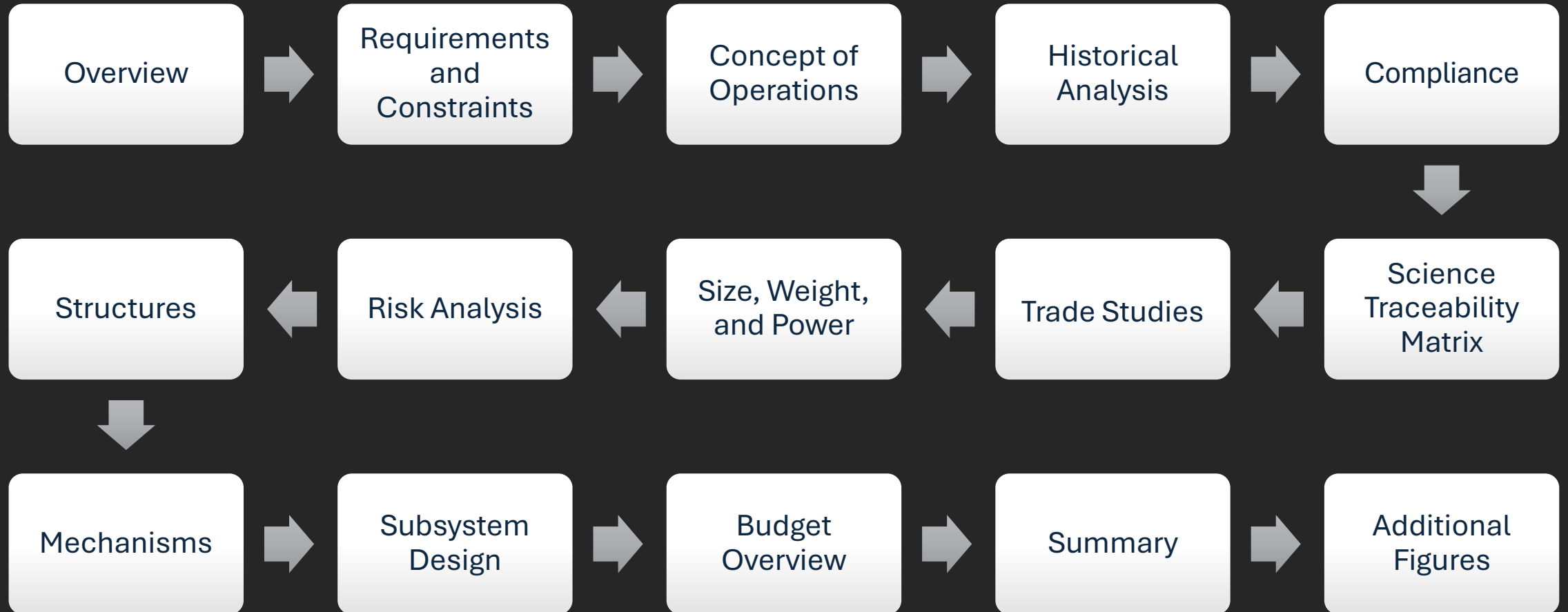
Preliminary Design Review

H.E.I.M.D.A.L.L.

Heliophysics Earth Interface and Mars Data
Acquisition Long-range Link



Agenda



Mission Statement

Provide future missions, both crewed and unmanned, a safe route to the Moon and Mars by detecting and characterizing Helio physical transients and ambient solar wind structures along the Sun-Mars transit corridor to quantify radiation, plasma, and magnetic field hazards while reducing communication blackouts with Earth.

Mission Objectives

MI-2

SE-2

Decadal Survey (to serve humanity)

"Analyze the space environment to project its future changes"

"Develop tools and products to issue space weather warnings and forecasts and to safeguard activities at and beyond Earth"

Mission Success Criteria

- To be successful HEIMDALL shall:
 - Minimize communications blackout duration between Earth and Mars during conjunction.
 - Contribute to the improved understanding of space weather phenomena by detecting and characterizing solar activity
 - Provide a warning service for solar radiation events that could affect satellite systems or the safety of the crew

Requirements and Constraints

AIAA Requirements
The satellite shall provide scientific instruments to support our understanding of the local cosmos, including the ability to monitor and detect solar and space weather phenomena, and enable the capability to provide early warning to crewed missions.
Ability to provide line of sight between the Earth and Mars to enable near constant communications capability during the REP specified mission period.
The mission design shall provide satellite and system analysis to show the capability for line-of-sight communication between Earth and Mars for the full Earth-Mars synodic cycle between 2036 and 2049.
AIAA Constraints
The cost of the mission and capability development in support of its activities shall not exceed \$400 Million US Dollar (in FY25), including development, hardware, launch, and operation cost of the mission through the primary mission phase.
The mission should launch no later than December 31, 2035, with the system designed to operate in the late 2040's

Space Weather Requirements
The payload instrumentation shall provide measurements of coronal mass ejections (CMEs).
The payload instruments shall measure the projection of CMEs and provide an estimation of its direction with a predetermined point accuracy to generate a space weather forecast model.
The payload instruments shall measure the interplanetary magnetic field vector in the standard Radial-Tangential-Normal coordinate frame.
Space Weather Constraints
The payload instruments will measure CME's acceleration over a range of 500 km/s ² to 3,000 km/s ² .
The payload instruments will measure the angular width and position angle of CMEs to an accuracy of $\pm 1^\circ$.
The payload instruments will provide an estimated arrival time at targets (Earth, Mars, Moon, and satellite traveling within these bodies) with an accuracy window of ± 6 hours.
The payload instruments will measure solar electron fluxes within a range of 100 keV to 15 MeV.
The payload instruments will measure proton and ion fluxes across an energy range of 10 MeV to 500 MeV.
The payload instruments will characterize lower energy seed populations at energy levels between 10 keV to 500 keV.
The payload instruments will measure magnetic field magnitudes over a range of ± 25 nT.

*Science payload was chosen based on achieving these R&Cs

Communications Requirements

The satellite shall provide a clear line of communication between Earth and Mars preventing lapses in communication.

The payload instruments shall provide position, navigation, and timing services for future crewed missions to Mars.

The satellite shall maintain communication functionality during nominal satellite operations and expected environmental conditions.

The communications hardware shall meet thermal limits and radiation tolerance requirements for the mission-specific environment.

The satellite shall operate in an approved deep space communications band (X-band) and comply with licensing requirements .

Communications Constraints

The satellite must be able to communicate with up to a range of 400 million kilometers.

The satellite must maintain antenna pointing accuracy sufficient to support high-gain communications with the surfaces of Earth and Mars, respectively.

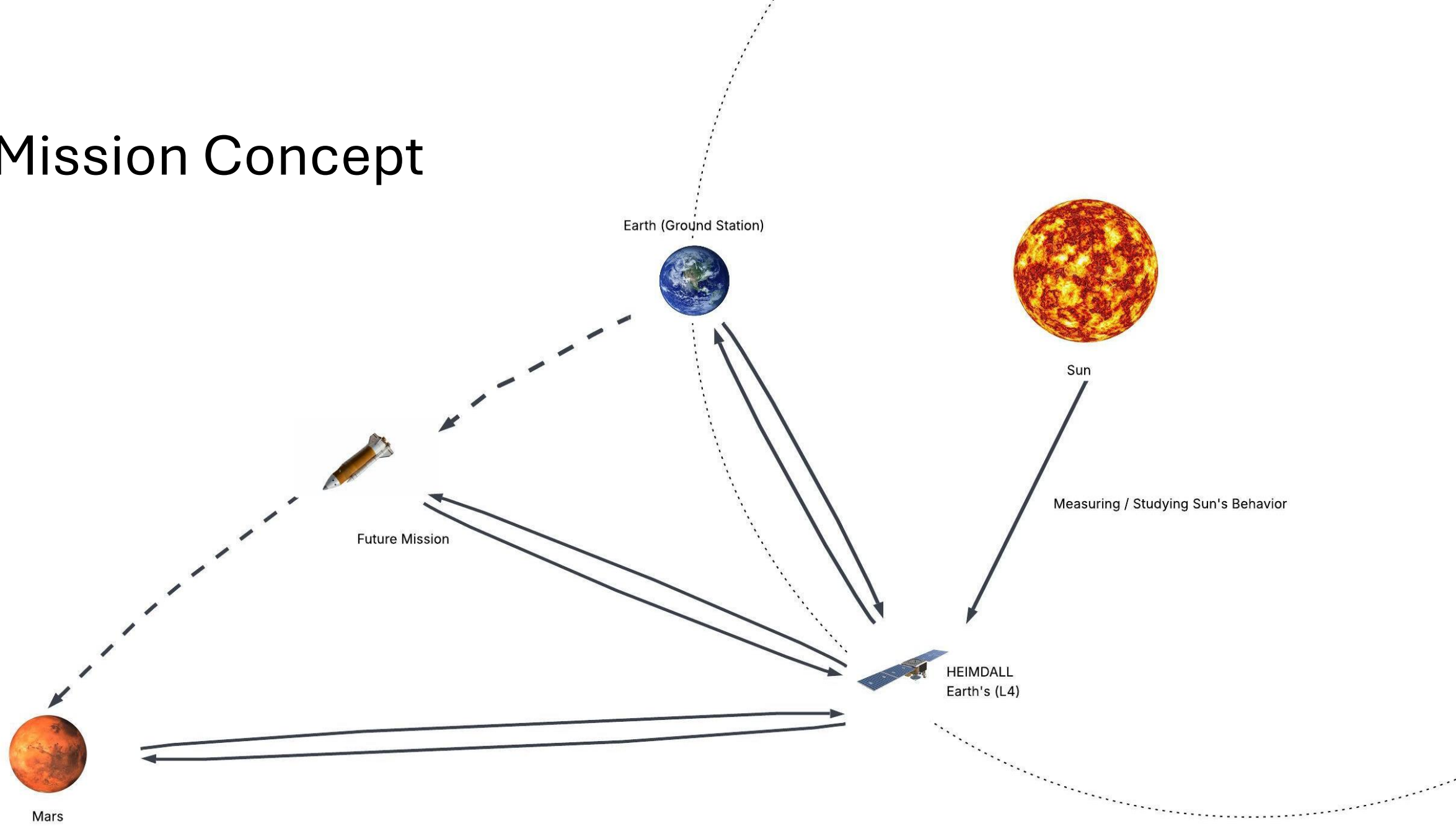
The satellite must be compatible with the intended ground network in modulation, coding, and protocols.

The communications system will operate within the satellite power budget, including peak transmit modes.

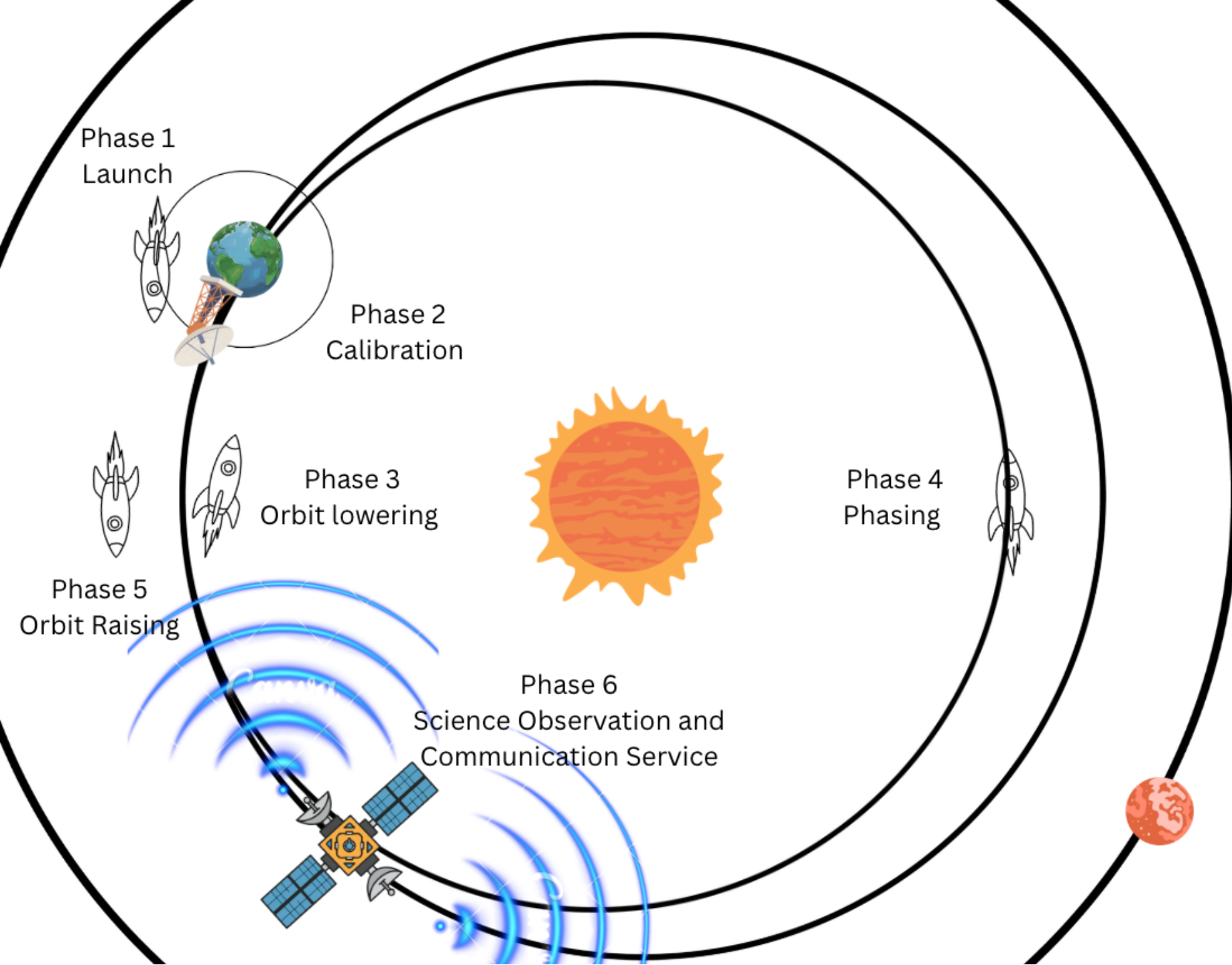
The communications system must meet satellite mass and volume allocations.

Concept of Operations

Mission Concept



Mission Schedule



Phase 1 (Sep 2035):

- SpaceX Falcon 9 (option)
- SLC-40 (Cape Canaveral)

Phase 2:

- Calibration and getting ready for the next maneuver at 300 km
- 5 days

Phase 3 (Oct 2035):

- $\Delta v_1 = 3.2366 \text{ km/s}$
- 9.31 degrees behind Earth velocity vector

Phase 4:

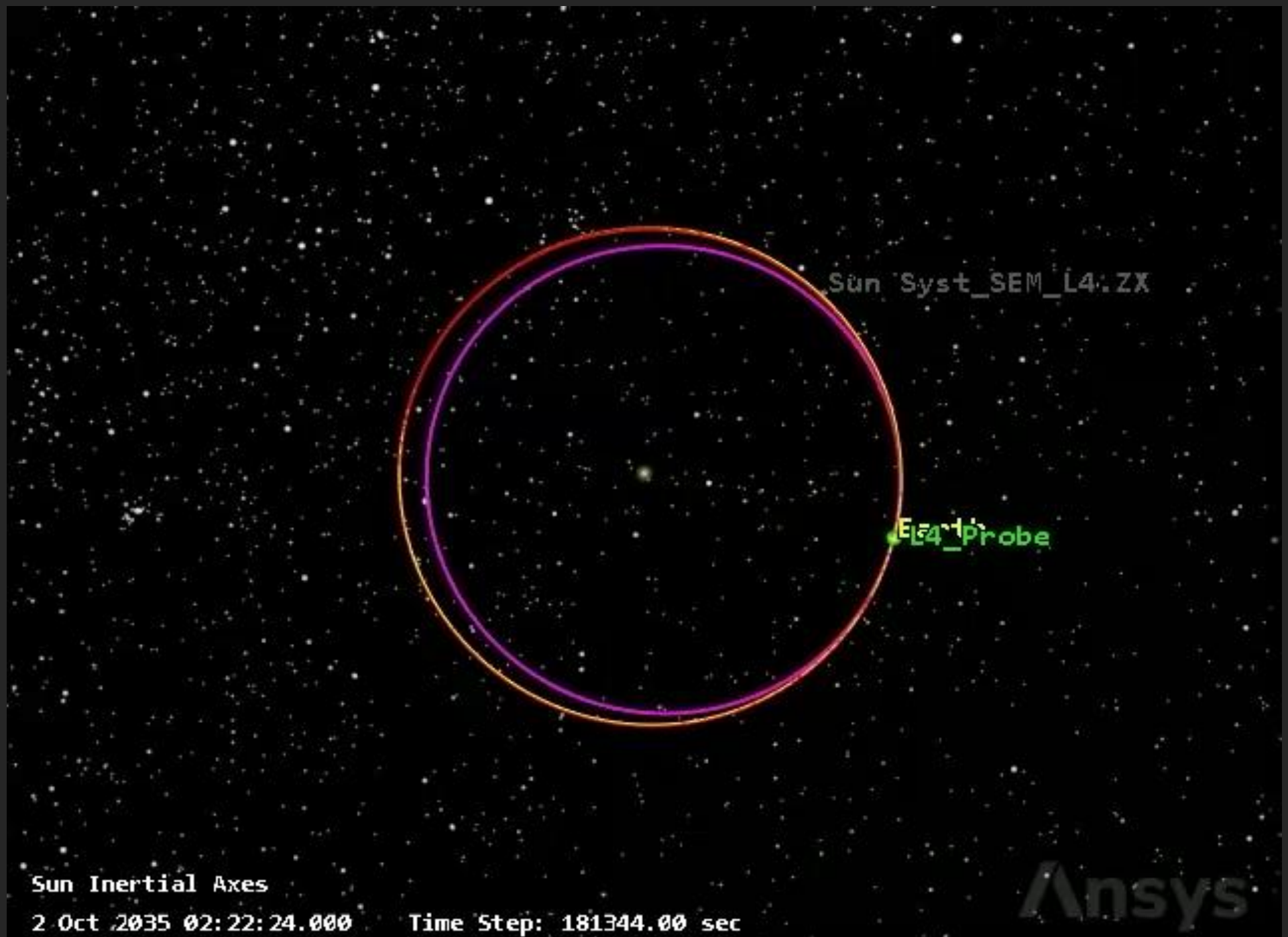
- Coasting for 668 days

Phase 5 (Aug 2037):

- Arrival at L4
- $\Delta v_2 = 0.8149 \text{ km/s}$
- Separation from upper stage

Phase 6:

- Operate at L4
- Heliophysics observations
- Communication service between Earth and Mars
- 12 years



Historical Analysis

Historically Relevant Missions

- **Mars Reconnaissance Orbiter (2005)** - Long range RF comms
- **MAVEN (2013)** - Large data packet burst transmission
- **Parker Solar Probe (2018)** - Heliophysics measurement
- **ESCAPADE (2025)** - Long range RF comms, launch vehicles
- **JPL PSYCHE (2023)** - DSOC, hybrid comms
- **IMAP (2025)** - Heliophysics measurement
- **OSIRIS-REx (2016)** - Deep space launch, long range RF comms
- **STEREO (2006)** - Heliophysics measurement
- **ESA Vigil (Planned)** - Lagrange Placement, Heliophysics measurement

Feasibility

Expense

New Frontiers

Venus or Mercury
Lagrange points



Trojan Asteroid
Op. Location



Earth L4 Position



Earth L5
Position



Objective

Feasibility

Expense

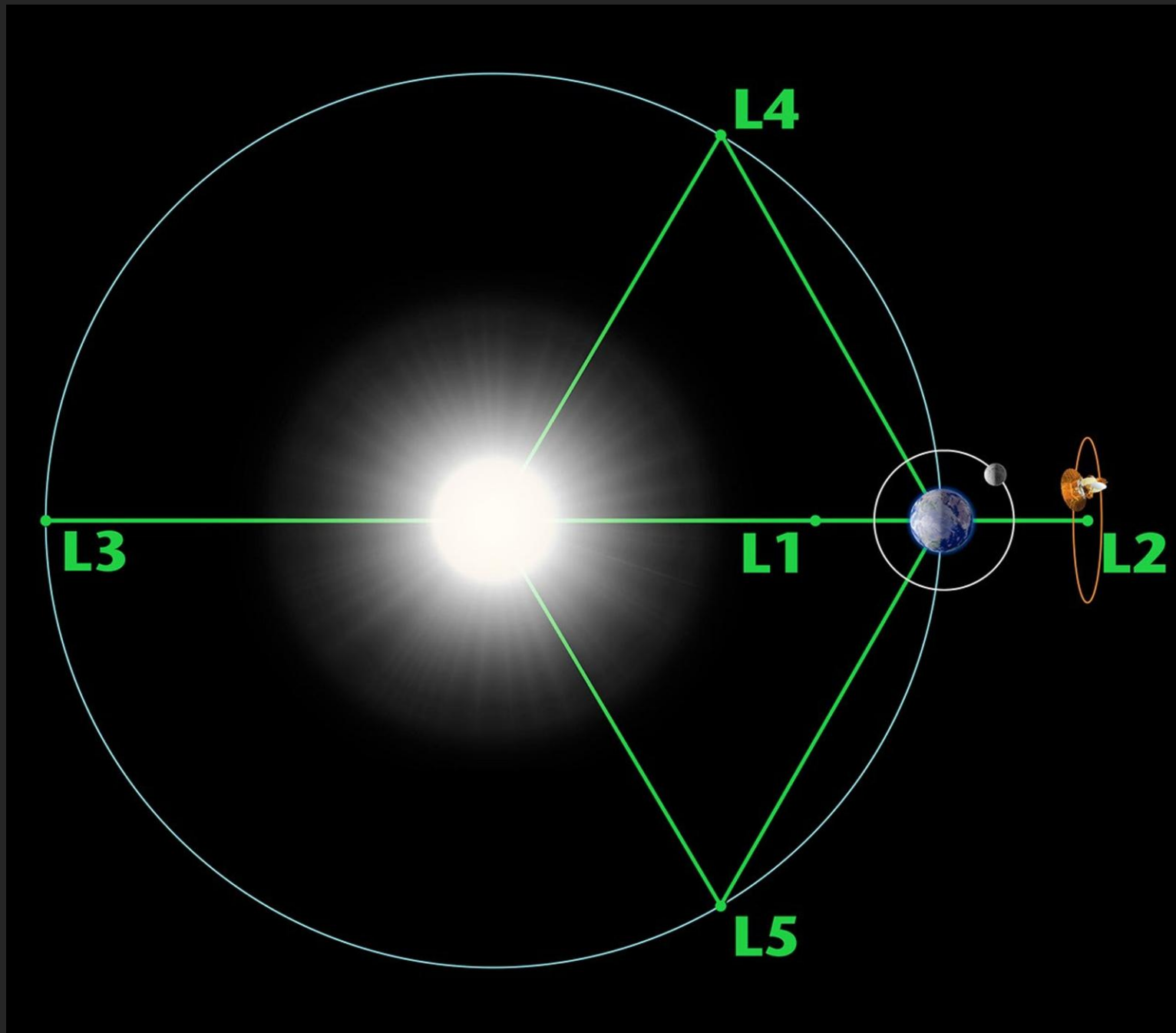
Novelty

Trojan Asteroid
Op. Location

Venus or Mercury
Lagrange points

Earth L5
Position

Earth L4 Position



Science Traceability Matrix

Science and Communication Objective	Measurement Requirements		
	Observables	Parameters	Instrument Type
Detect, characterize, and provide timely warning of heliophysical transients and ambient solar-wind structures along the Sun-Mars transit corridor to quantify radiation, plasma, and magnetic-field hazards to Mars-bound crews and vehicles.	Coronal Mass Ejection (CME) Detection	White light coronal brightness evolution and CME front morphology; Leading edge speed; Acceleration; Angular width; Propagation direction; Brightness/mass determination; Initial expansion rate.	White-light Coronagraph
	CME Propagation and Projection	White light elongation time track of the CME leading edge; Radial distance vs. time; Speed; Propagation long term acceleration/deceleration; Direction relative to Mars transit line; ETA to points of interest; Arrival speed	Heliospheric Imager
	Solar Energetic Particles (SEP)	Electron, proton, and ion differential flux versus energy and time; SEP onset time; Peak SEP flux; Expected event duration; Spectral hardness; Radiation storm severity.	Energetic particle telescope / spectrometer

Equipment selection made in partnership with the trade study to provide a more complete scientific suite that meets mission requirements

Trade Study

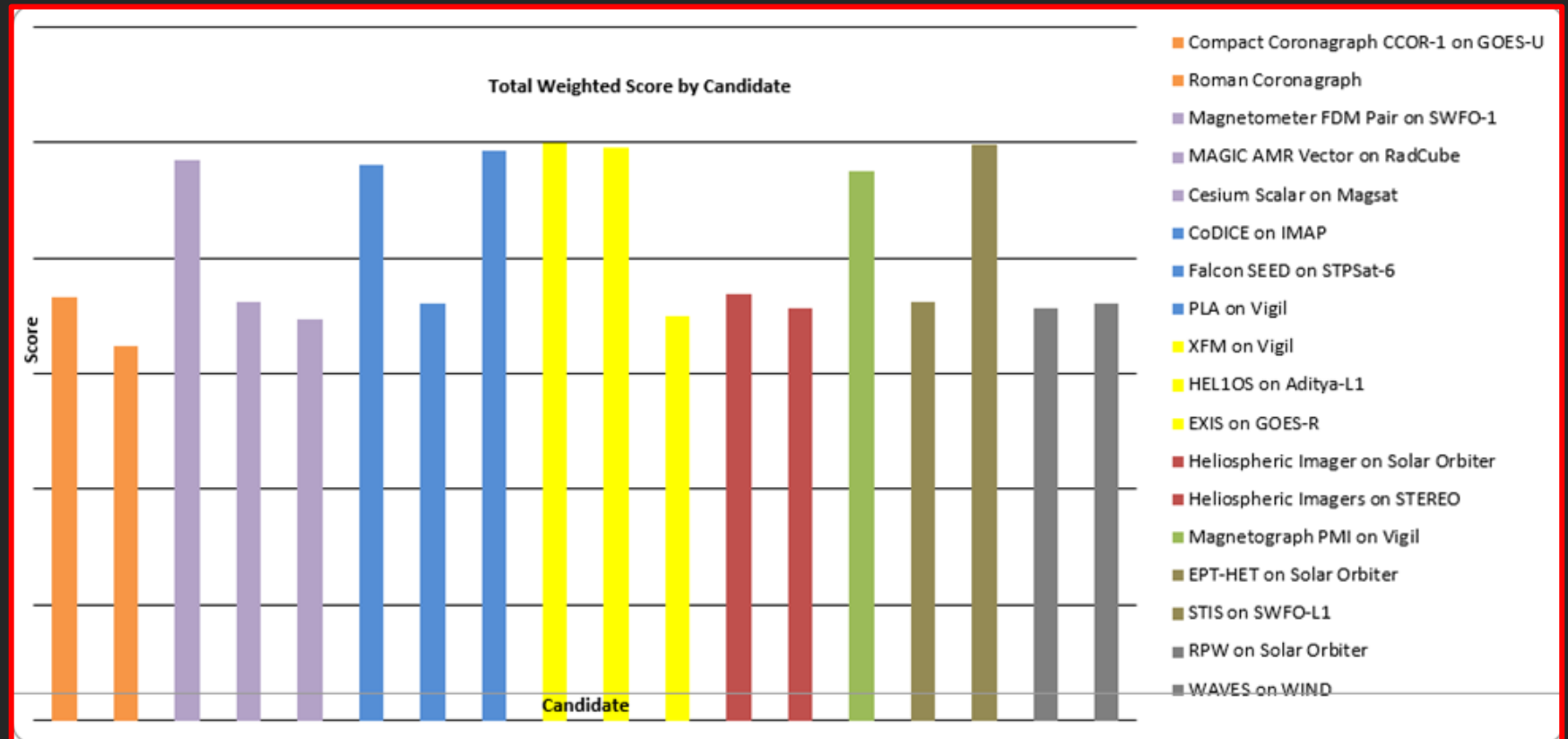
Scientific Payload Trade Study

Top Performing Instruments:

- CCOR-1
- MAG FDM
- PLA
- XFM
- SOLOHI
- PMI
- STIS
- WAVES

- Instruments were chosen based on highest ratings and alignment with our science R&Cs (slide 6)

LEGEND	
	Coronagraphs
	Magnetometers
	Plasma Analyzers
	X-Ray Flux
	Heliospheric Imagers
	Magnetograph
	Energetic Particle Telescopes/Spectrometers
	Radio Burst Monitors



Size, Weight, and Power

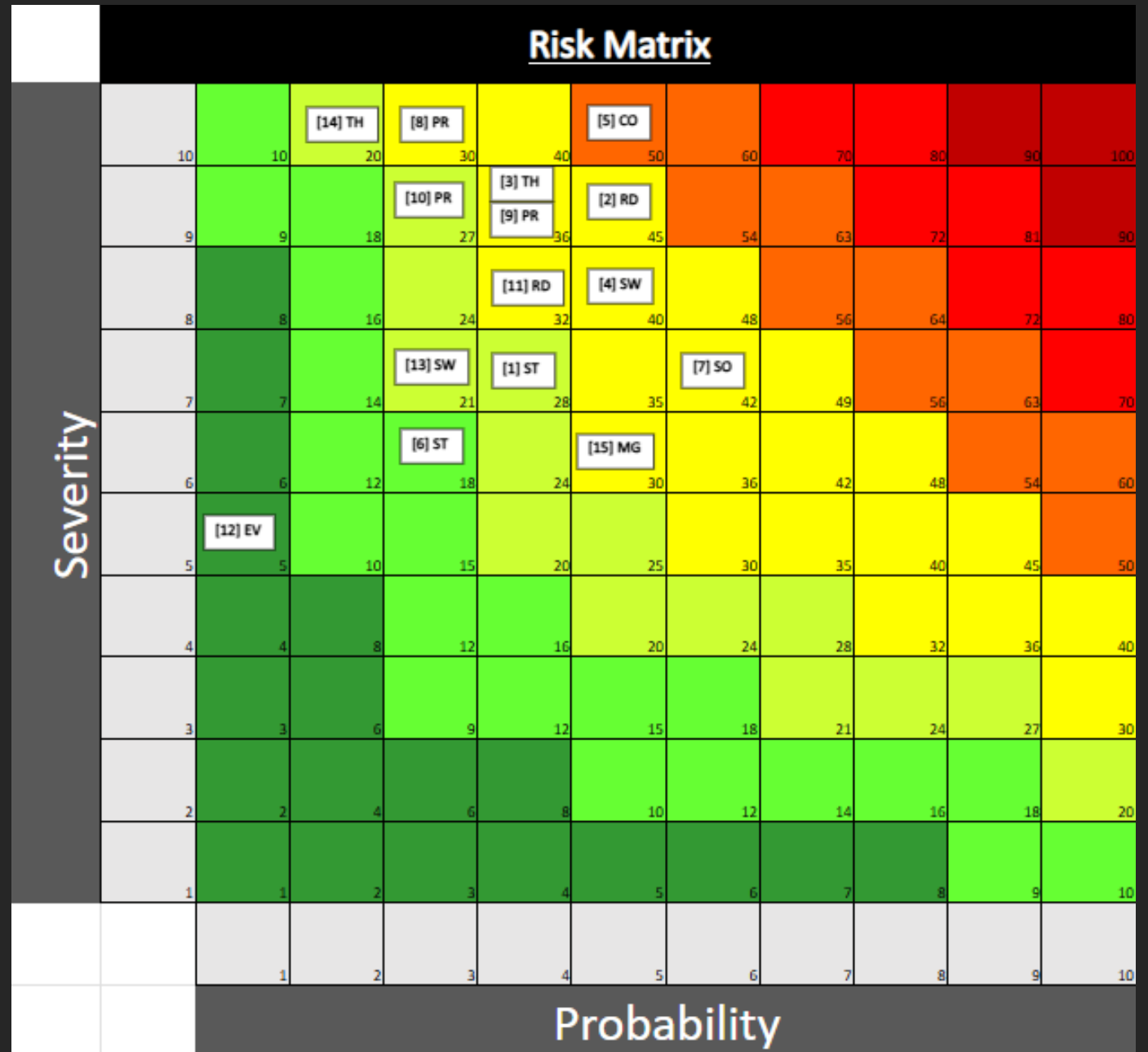
SWaP

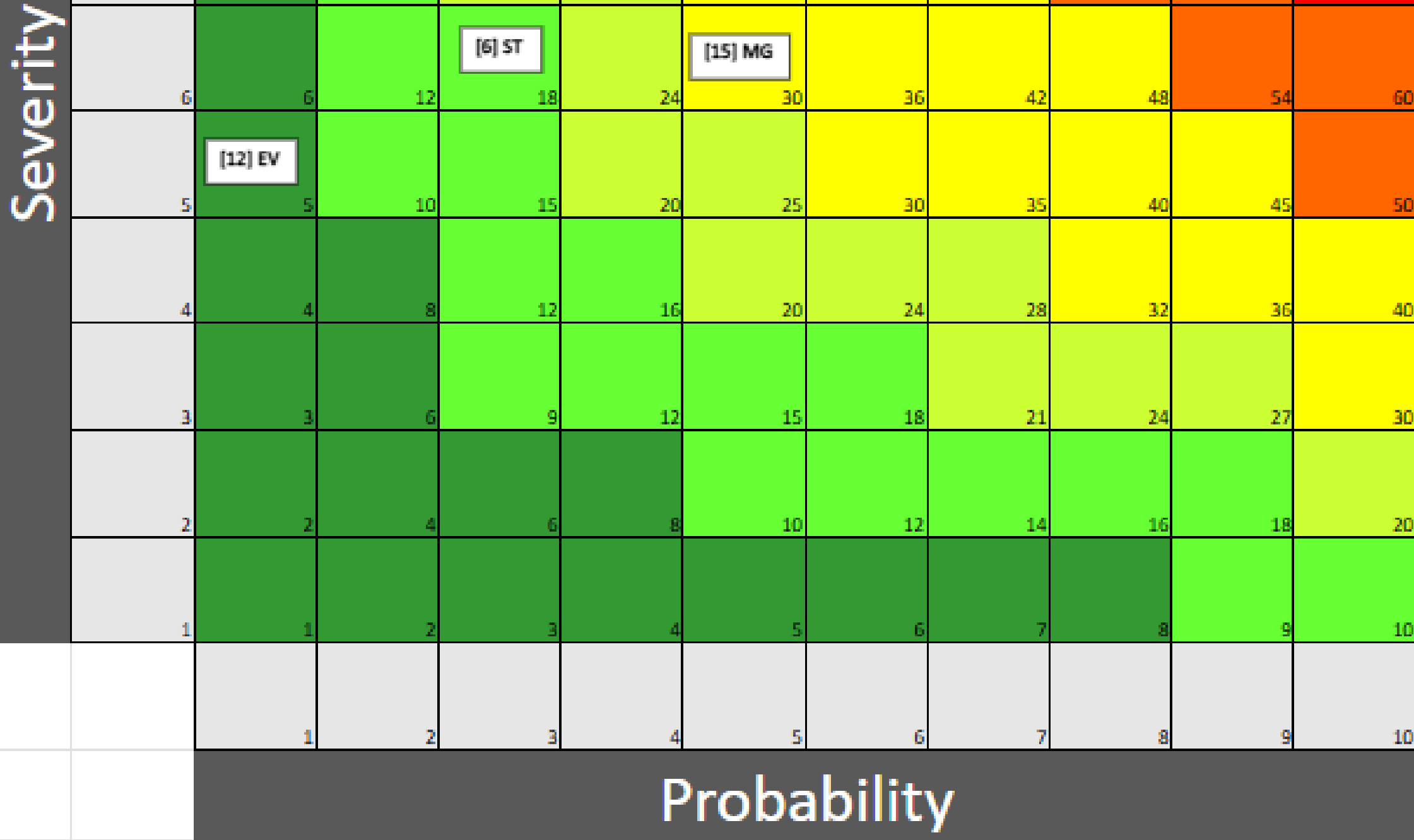
- Enable early Space Weather Detection
- Establish and Maintain Consistent Comms between earth and Mars.
- See slides 77-78

Risk Analysis

Risk Analysis

- Measuring 15 different hazards
- Trend: moderate probability but high severity towards mission failure





Risk Matrix

[illegible]

ID	PR	ST	CO	SO	TH	EL	MG	RD	SW	EV
Category	Propulsion	Structure	Comms	Science Observation	Thermal	Electrical	Management	Radiation	Software	Environment

Impact	Risk Level	Mitigation Strategy
EM interference could disrupt signal clarity for incoming transmissions, jeopardizing critical comms and s/c instructions which may be time-sensitive or unrepeatable.	42	EM shielding of critical systems, booms distancing antennae from EM-emitting components, insulating material shielding high-voltage systems
Degradation or failure of spacecraft electronics and sensors, reducing mission lifetime and scientific data return.	45	Radiation shielding, automatic safe-mode planning, utilizing solar data for predictive modeling of radiation spikes.
Loss or interruption of communication with Earth could prevent transmission of scientific data and telemetry and limit mission operations.	50	MGA/LGA backup comms, redundancy in comms hardware, ability to substitute Mars-side/Earth-side systems in case of emergency.

Structures Subsystem

Requirements and Constraints

Structural Subsystem Requirements
Being located at L4 will expose HEIMDALL to extreme radiation and cosmic rays that the structural material will have to survive for a mission duration of 15 years.
The HEIMDALL bus must survive the impact and vibrations during launch and launch vehicle separation; the structural materials have to be able to absorb as much energy so that the internal components receive as little damage as possible.
The structural subsystem will require materials with high fracture toughness for the outside layers since these will be exposed to all sorts of space particles, small meteoroids, etc.
Structural Subsystem Constraints
The structure design must fit inside the launch vehicle and have a destined attachment support.
The materials must be on the lighter scope as we all maintain a relatively accessible price and great manufacturing.
In order to maintain good communication between Earth, Moon, and Mars as well as having a direct line of sight to the Sun, the materials chosen must have a good cryogenic performance to avoid changes in dimension when experiencing different temperature ranges.
Temperature can range from -100 °C to 150 °C and the materials chosen must maintain good mechanical properties at these temperature ranges.

Material Selection

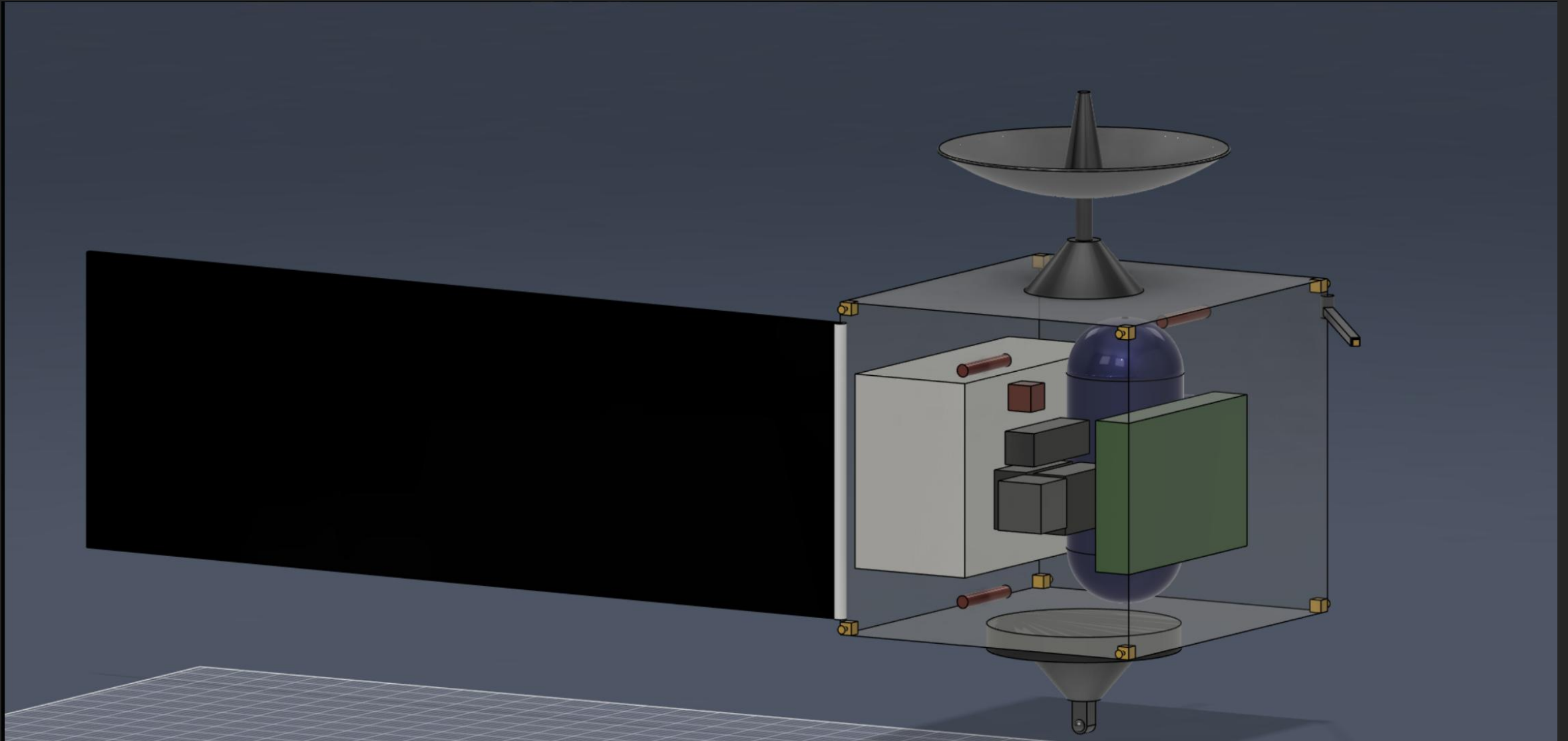
Material	Aluminum 6061-T6	Aluminum- Lithium 2195	Ti-6Al-4V	CFRP
Density (g/m ³)	2.7	2.56	4.43	1.6 – 1.9
Tensile Strength (MPa)	210	570	950 – 1380	1500
Thermal Expansion	High	High	Excellent	Excellent
Cryogenic Performance	Good	Excellent	Excellent	Good
Radiation Resistance	Excellent	Excellent	Excellent	Poor
Pyrotechnic Shock Tolerance	Good	Excellent	Excellent	Vulnerable
Manufacturability	Excellent	Good	Complex	Complex
Raw Material Cost (USD/KG)	\$4 – 7\$	\$15 – 25\$	\$40 - \$150	\$90 - \$200

Material Selection



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Manufacturability	Excellent	Good	Complex	Complex
Raw Material Cost (USD/KG)	\$4 – 7\$	\$15 – 25\$	\$40 - \$150	\$90 - \$200

Mechanisms



This is an early-stage design. There will be another solar panel on the right side creating a more symmetric body.

Thermal Control Subsystem

Thermal Subsystem Requirements & Constraints

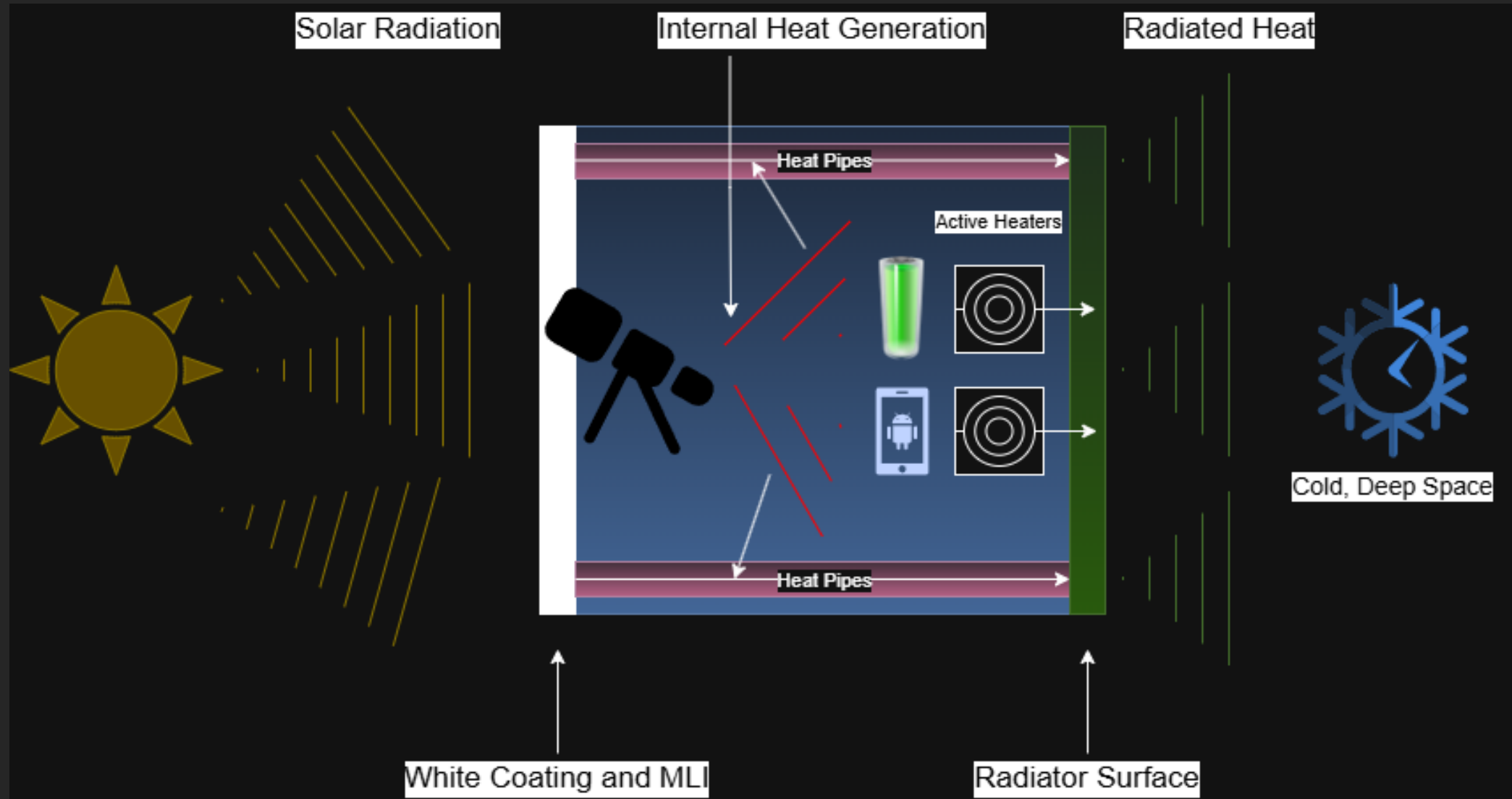
Thermal Subsystem Requirements
The thermal subsystem shall maintain all internal components within their specified operational temperature during all phases of the mission. The temperature should be between -30 °C and 130 °C.
The thermal subsystem shall radiate collected internal bus heat into the space environment via radiators mounted on the on the anti-Sun face of the satellite.
Thermal Subsystem Constraints
To keep power budget low, the thermal subsystem architecture prioritizes passive thermal control methods before implementing active control methods.
All exterior thermal coatings and layers must be rated to survive 14 years of deep space ultraviolet and proton radiation.
Batteries, some science instruments, and communications equipment have narrow temperature ranges, -5 to 25, -35 to 5, and -10 to 50 degrees Celsius, respectively.

*Report will note the instruments that are most important that the thermal system is based on

Thermal Control Subsystem

Thermal Mechanism	Passive or Active	Description
Multi-Layer Insulation (MLI) Blankets	Passive	• 10-20 layers of Aluminized Kapton and Dacron mesh spacers • Emissivity: 0.01-0.03 • Heat Flux Reduction: 70-90% • Mass: 0.1-0.3 kg/m²
AZ-93 Zinc Oxide-based White Coating	Passive	• Absorptivity: 0.17-0.20 • Infrared Emissivity: 0.90-0.92 • Mass: negligible
Aluminum Heat Pipes	Passive	• Mass: 0.33 kg/m (1.27 cm diameter)
AZ-93 High-Emissivity Coated Aluminum Radiator Panels	Passive	• Emissivity: ~ 0.9 • Heat Rejection Capability: 300-400 W/m² • Mass: 0.2-0.6 kg
Minco Polyimide Thermofoil Kapton Film Heaters	Active	• Power Density: 0.8-5.1 W/cm² • Temperature Capability: -200 to +200 degrees Celsius • Peak Power: 5-10 W (1-5 W average) • Mass: < 50 g

Thermal Design Model



Attitude Determination and Control System (ADCS)

ADCS Subsystem Requirements & Constraints

ADCS Requirements

The ADCS system shall maintain low drift and jitter to support the pointing and stability requirements of HEIMDALL's onboard scientific instruments and communication relay needs.

The system shall provide accurate three-axis attitude determination and control throughout HEIMDALL's operational lifetime.

The ADCS system shall support all five operational modes of HEIMDALL: science mode, communications relay mode, safe mode, acquisition mode, and momentum unloading mode.

The subsystem shall provide safe mode sun pointing capability to maintain power to the satellite.

ADCS Constraints

The ADCS system must operate in the Earth-Sun L4 environment, where solar radiation pressure is the dominant external disturbance.

ADCS Overview

- The ADCS determines and maintains HEIMDALL's attitude throughout the mission.
- MODES
 - **Science** – high accuracy, low jitter pointing for the coronagraph, heliospheric imager, and telescope/spectrometer.
 - **Communication Relay** – maintains orientation for antenna alignment during relay operations.
 - **Safe** – coarse Sun pointing to maintain power and thermally safe conditions.
 - **Acquisition** – determines and stabilizes satellite attitude after disturbances or mode changes.
 - **Momentum Unloading / Slew** – reorients the satellite and unloads stored reaction wheel momentum.

ADCS

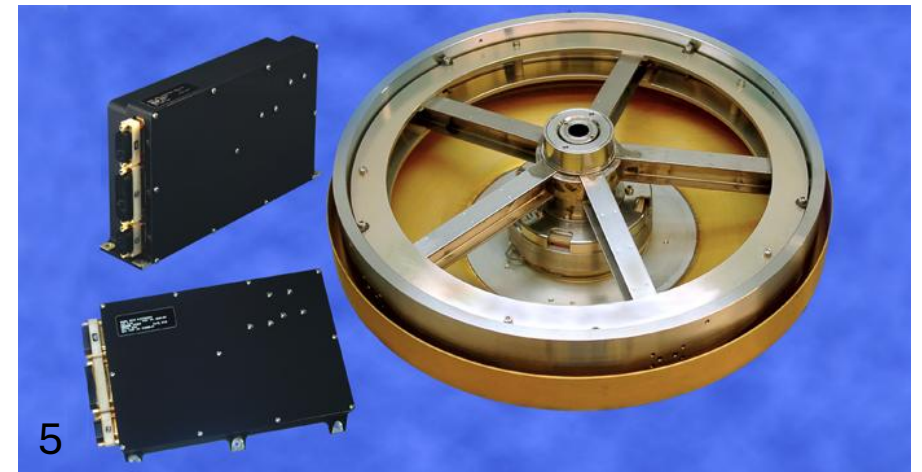
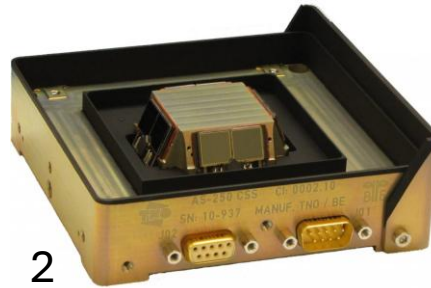
Variable	Description	Value
F_s	Solar flux	$1372 \frac{W}{m^2}$
C	Speed of light	$3 \times 10^8 \frac{m}{s}$
A	Illuminated surface area	$2.38 m^2$
q	Reflectance factor	.6
i	Sun incidence angle	0 deg
d	Offset of center of pressure and center of gravity	.5

$$F_{srp} = \frac{F_s}{c} A(1 + q) \cos i \approx 1.74 \times 10^{-5} (N)$$

$$T_{srp} = F_{srp} d \approx 8.7 \times 10^{-6} (N \cdot m)$$

Preliminary ADCS Components

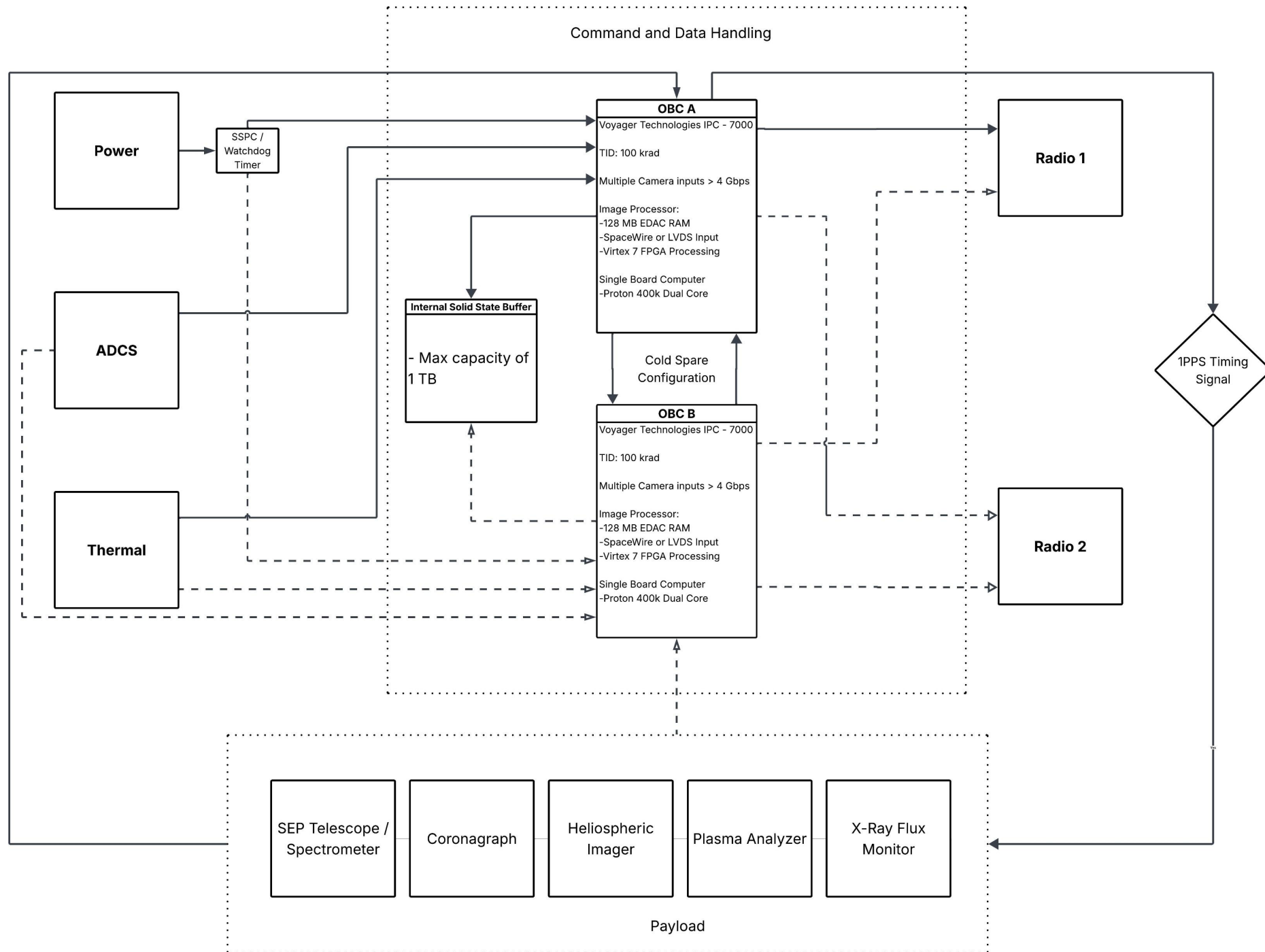
1. LR-450 IMU 1x
2. Coarse Sun Sensor 6x
3. ASTRO CL Star Tracker 2x
4. MONARC-5 8x
5. RDR 68 4x



Command & Data Handling (C&DH)

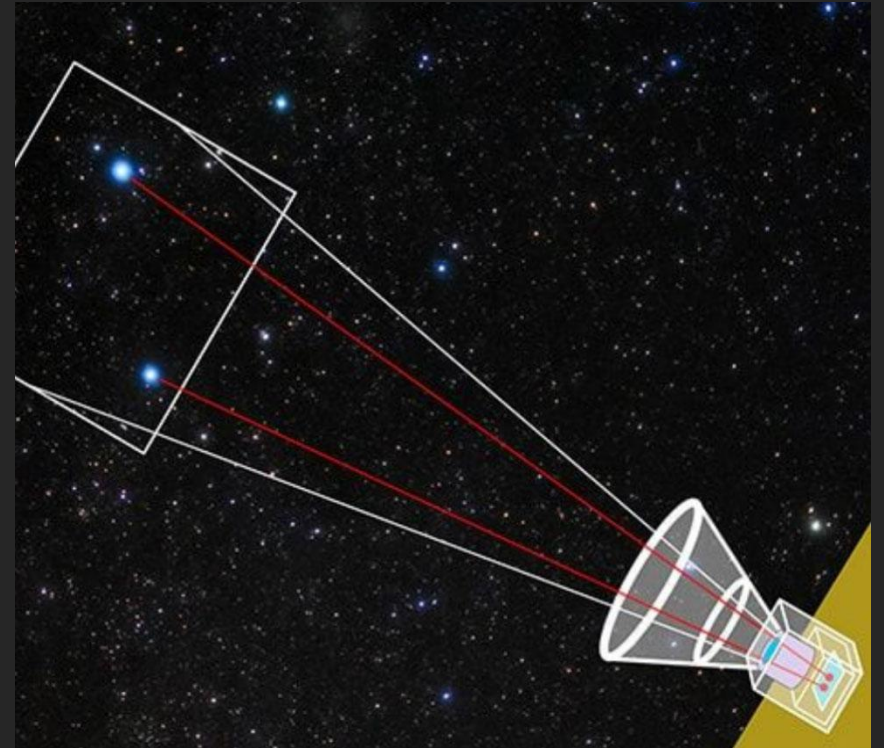
C&DH Requirements and Constraints

Command & Data Handling Requirements
The system shall be capable of applying time-synchronization commands from the ground station during scheduled TT&C uplink windows.
The C&DH subsystem shall distribute a continuous 1 pulse per second hardware synchronization signal across the data bus to ensure timing accuracy between the payload and the TT&C subsystems
The OBC shall be able to process all the data collected and perform real-time JPEG compression on raw solar images prior to routing data to storage.
The OBC should count with a reliable redundancy system in the event of extreme solar radiation that might damage sciential components.
Command & Data Handling Constraints
To survive deep-space environment at Lagrangian point L4, all critical C&DH processing components must have a Total Ionizing Dose (TID) of at least 100 krads.

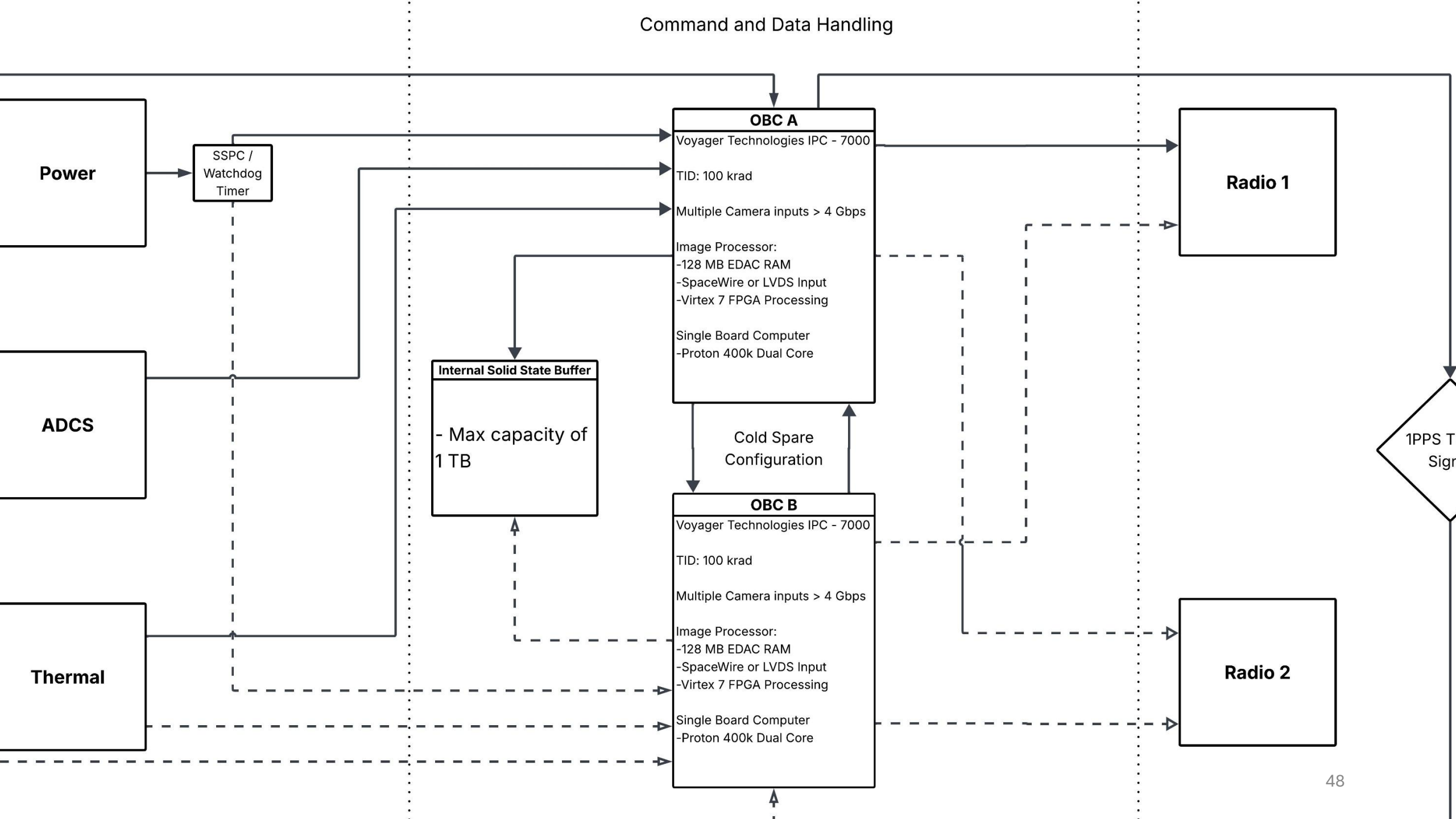


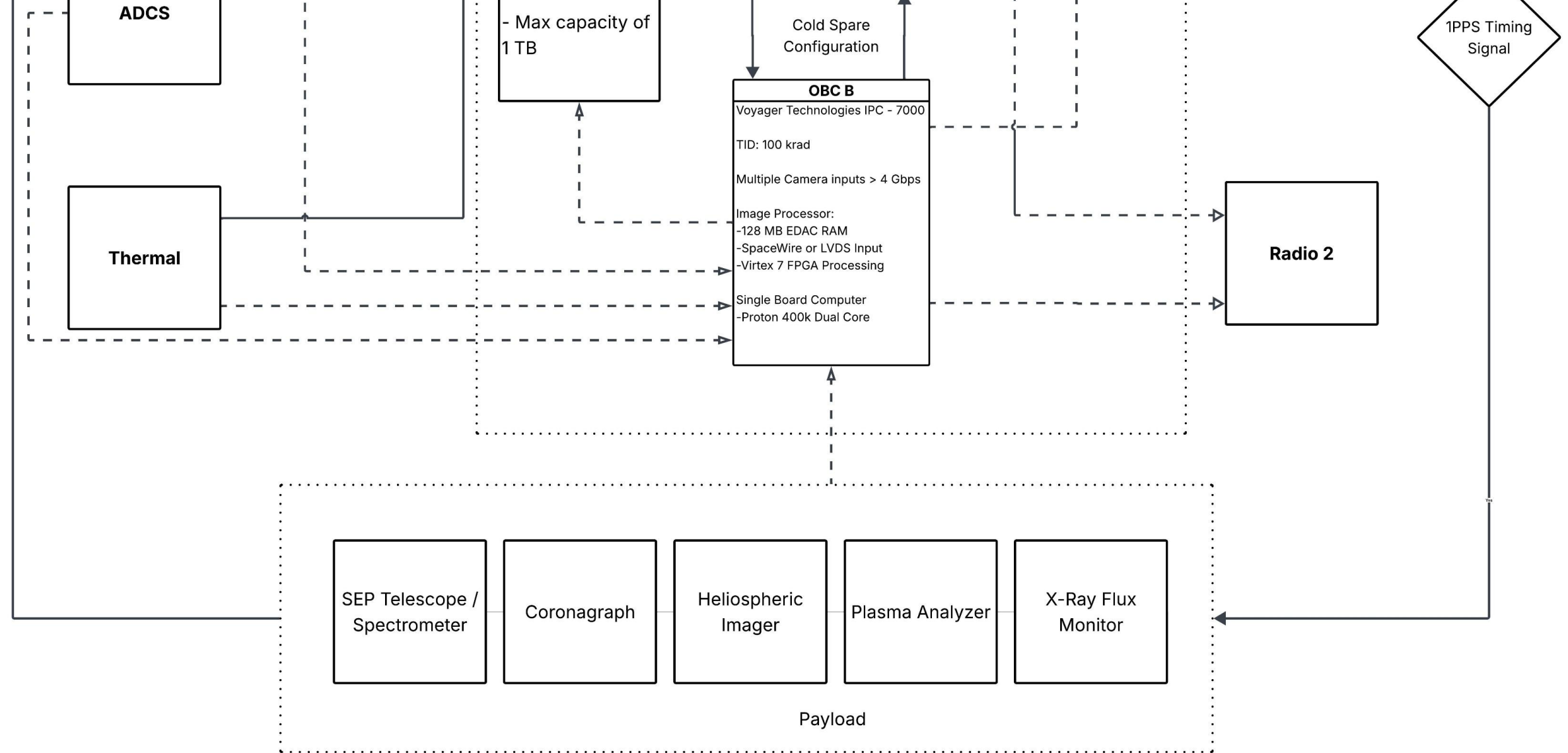
Telemetry and Tracking

- Constellation tracking
- Sun tracking
- X-band range finding via Isotropic burst

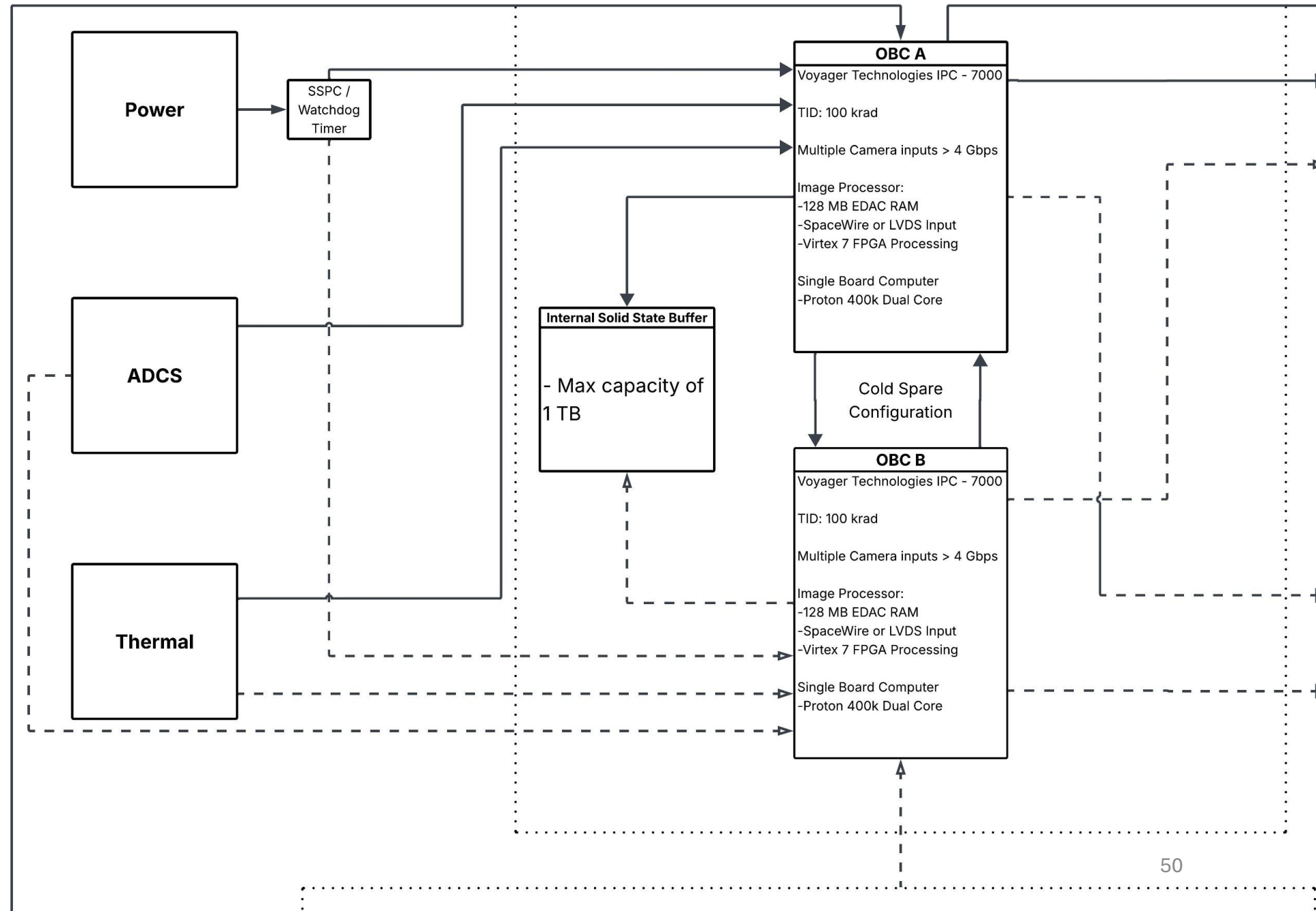


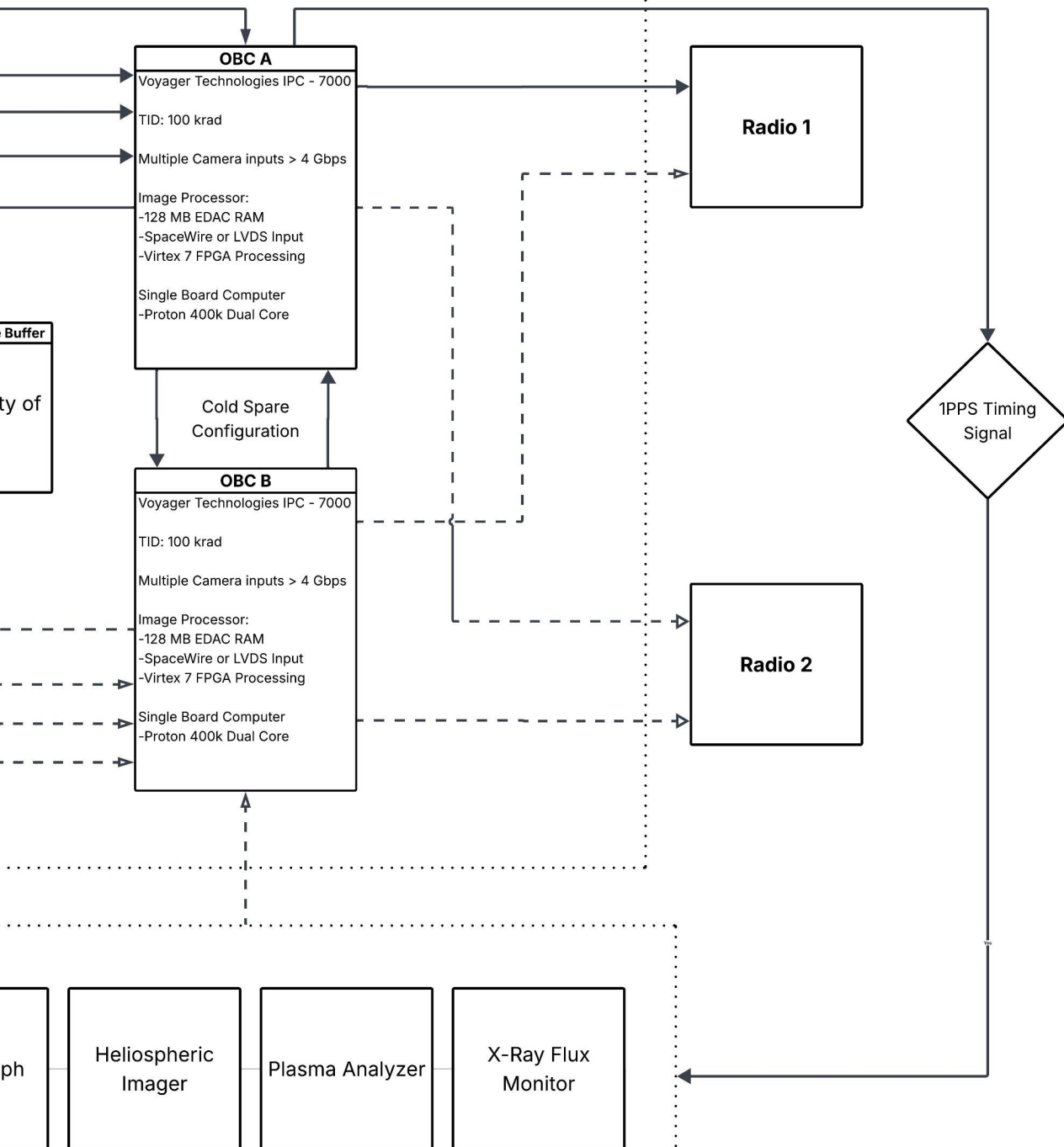
Command and Data Handling





Command and Data Handling

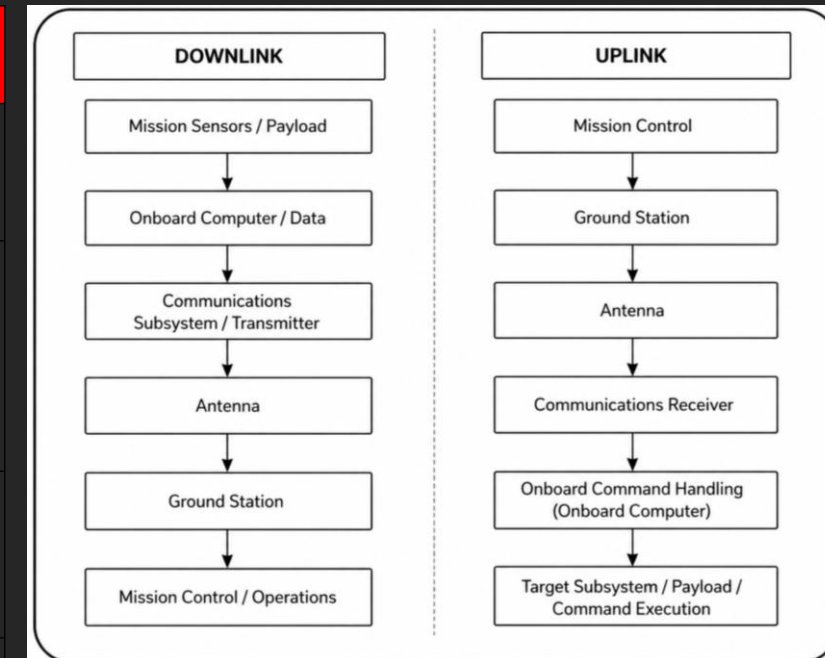




Communications

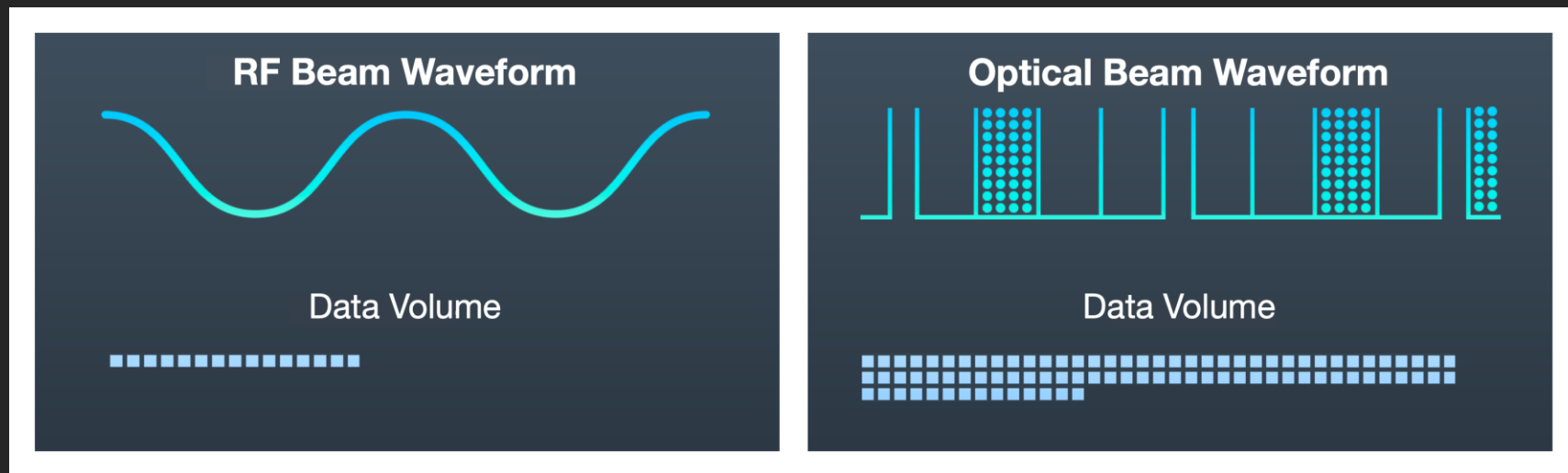
Communications Requirements

Req No.	Requirement Statement	Direction, Source - End User	Data Rate	Access/Delay	Availability/Reliability	Criticality
1	Science payload data shall be downlinked for mission analysis.	Downlink, Payload - Mission Control	2 Mbps / 0.9 G B per contact hr	Per pass/< 24 hr	High/High	Medium
2	Status telemetry shall be available $\geq$ 99% of contact time with a maximum latency of 10 minutes per pass.	Downlink, OBC/subsystems - Mission Control	5 kbps	Per pass/< 10 min	High/High	Medium
3	Fault telemetry shall be prioritized for response in the case of an anomaly.	Downlink, OBC/FDIR - Mission Control	Low, event-specific	Near real-time	Very high/Very high	CRITICAL
4	Critical commands shall be uplinked for emergency actions	Uplink, Mission Control - OBC	Very low	Minimal delay	Very high/Very high	CRITICAL
5	Payload/configuration commands shall be uplinked for instrument tasks.	Uplink, Mission Control - OBC	Low	During contact/< 1 hr	High/ High	Medium
6	Routine commands shall be uplinked for nominal mission operations.	Uplink, Mission Control - Payload/subsystem	Low	Scheduled contact/< 1 hr	High/High	Low
7	Relay communication via Mars crew assets shall be supported.	Uplink/Downlink, Mission Control $\leftrightarrow$ Spacecraft (via Mars relay)	Low – Medium 1 kbps – 1 Mbps	Relay delay $\leq$ 5 s	High/High	High



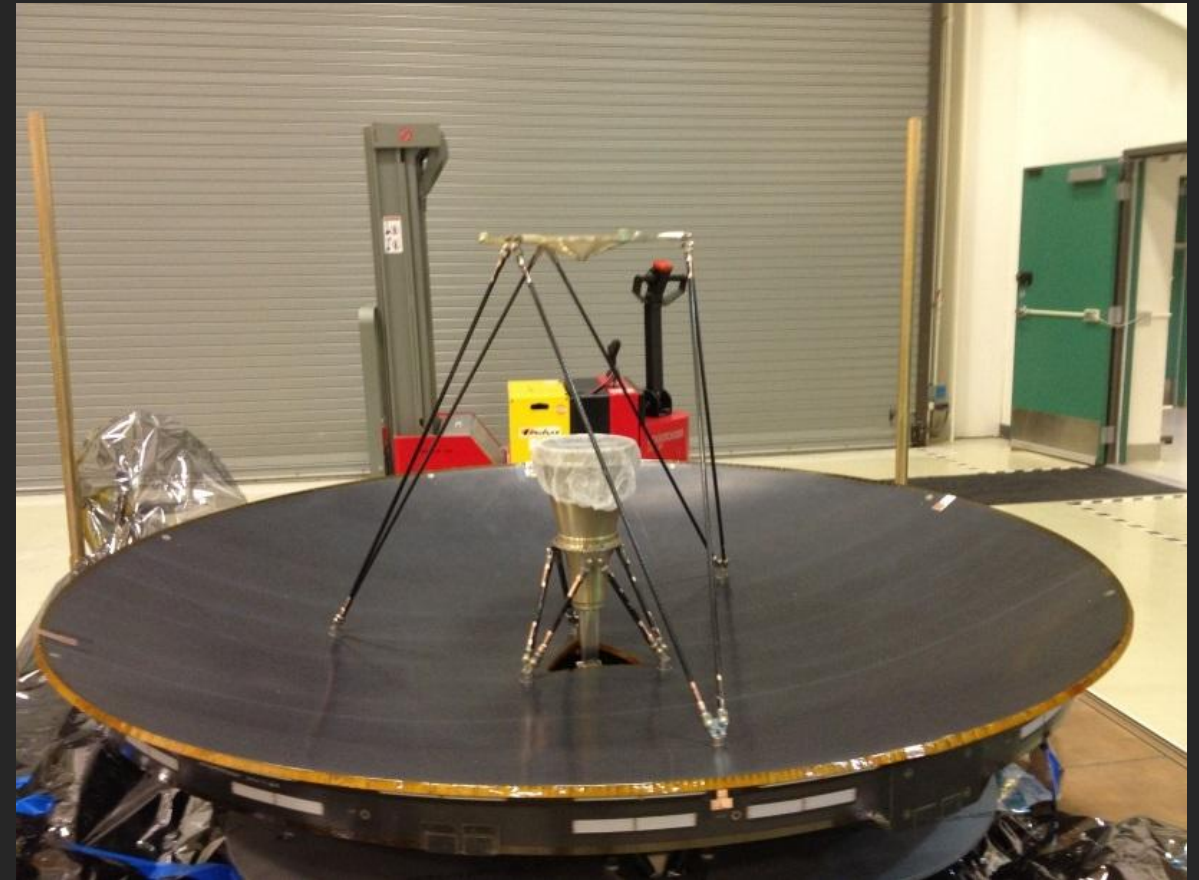
Communications Overview

- Hybrid system to provide redundancy; achieve high-bitrate high-reliability communications
- Maximize reliability by providing high-powered X-band RF system for one-way transmission
- Maximize bitrate by using Deep Space Optical Comms (DSOC); likely to become the standard for long range communication in future



X-band Radio Frequency System

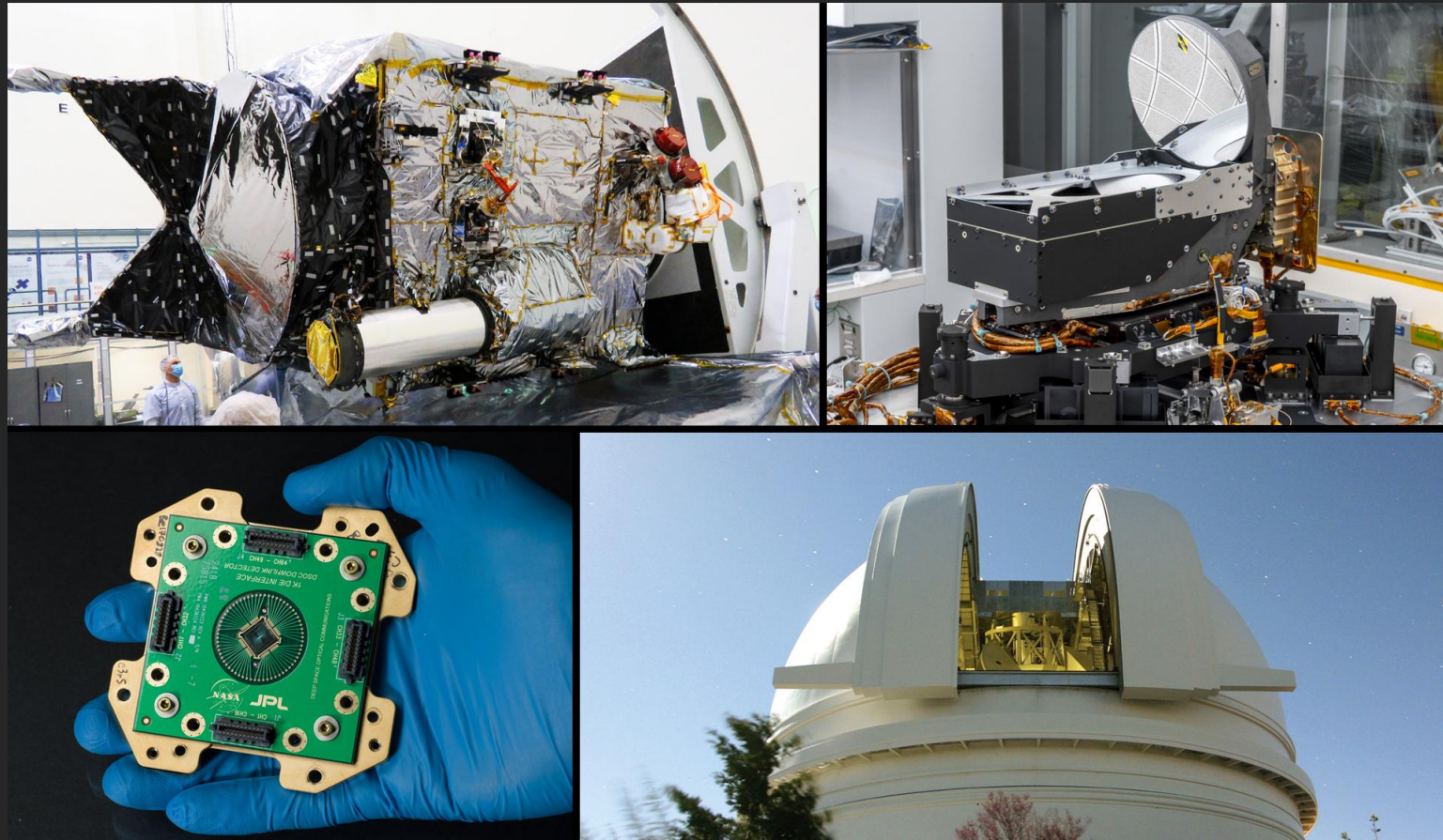
- Heritage concept with over 60 years of flight-proven performance at deep space ranges (1967)
- Receivable without need for modification at existing ground stations
- Low-bitrate, less demanding on pointing accuracy (± 0.1 deg)
- Relatively simple to manufacture and operate



Pictured: NASA OSIRIS-REx HGA

Deep Space Optical System (DSOC)

- First flight proven much more recently than RF (2023)
- Very high bitrate allowance, opens potential for video streaming, large data packets
- Extremely sensitive to vibration and pointing accuracy (± 0.001 deg)



Clockwise from top left: NASA JPL Psyche satellite, Onboard DSOC system, ground transmitting telescope, and photon counting chip

Portioning Data Streams

- **X-band:** reserved for scheduled scientific transmissions, health, telemetry of satellite; available for re-tasking to Communications Objectives
- **DSOC:** reserved for relay between Earth, Mars, and deep space missions requiring high bitrate; available for re-tasking to Science Objectives
- Redundancy of hybrid architecture allows for scientific objectives to remain achievable with DSOC failure, and communications objectives to be achieved in the event of X-band failure
- Also allows for simultaneous transmission in uplink, downlink, or bidirectional communications

Predicted Peak Data Rates at 1 AU

Link	Ground Station	Future Missions in Transit
X-band Uplink	1.8 kbps	Not Intended
X-band Downlink	180+ kbps	Not Intended
DSOC Uplink	1.8 kbps	25+Mbps
DSOC Downlink	25 Mbps	25+Mbps

X-band bitrates calculated via ground station parameters at Goldstone, CA for 1 AU range.

DSOC bitrates taken as referenced at 1-2 AU range during JPL Psyche DSOC demonstration.

Power

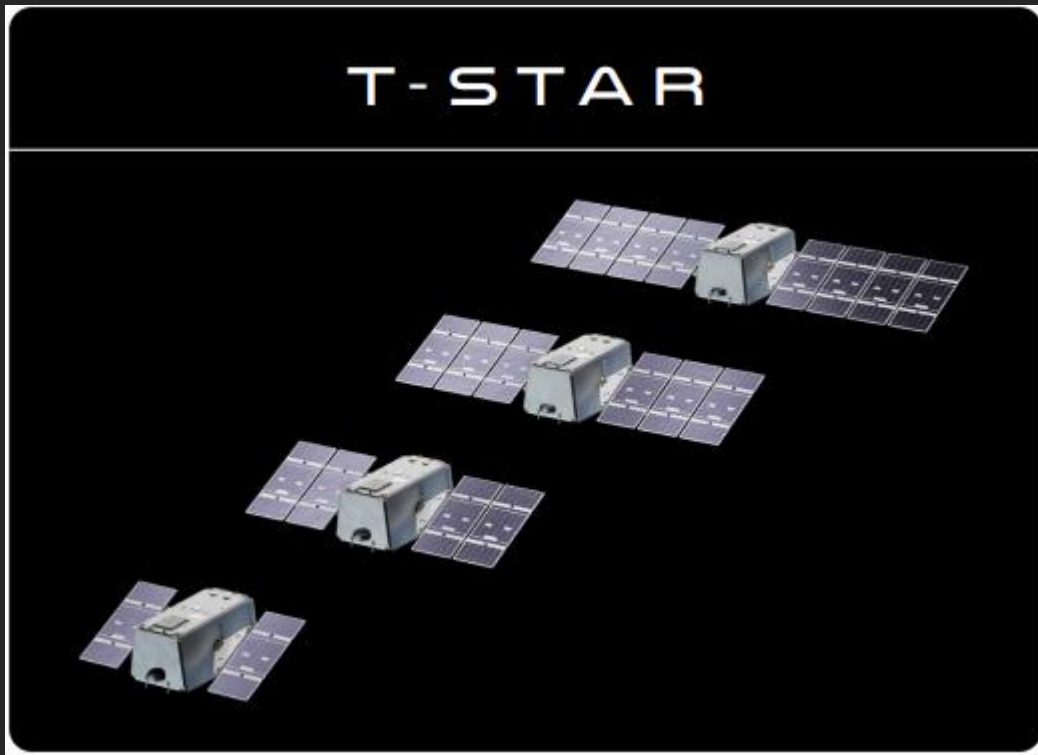
Subsystem	Daily Average Power Output (W)
Payload	136.01
ADCS	56.2
C&DH	40
Power	10
Propulsion	0
Structure	0
Thermal	2
Communications	149
Power margin (20%)	78.642
Total Power	471.852

Component (Type)	Component Average Daily Power Output	Theoretical Max Power Output	Station Keeping	Deployment
WAVES Heritage (Radio Burst Monitor)	8	8	0	0
PLA on VIGIL (Solar Wind Ion/Plasma Analyzer)	14	14	0	0
VIGIL Heritage X-ray Monitor (X-ray Flux Monitor)	30	30	0	0



Laser transmitter	4	4	0	0
Laser system	75	100	0	0
20% Power Margin	78.642	181.74	70.84	53.4
Power (W)	471.852	1388.15	425.04	320.4

See Full Table at Slide (72)



Solar Panels are built by Rocket Lab

After accounting for daytime generation efficiency, the required power is approximately **555 W**.

To ensure this requirement is met at the end of life (EOL), the corresponding beginning of life (BOL) power is calculated to **665 W**,

This translates to **~333 W per solar array wing**.

T-Star BOL Vmp and size options with wing power

Voltage Option	Substrate Size (mm)	Wing Power (Pmp, beginning of life, AMO 28°C and 0° angle of incidence)			
		1 Panel per Wing	2 Panels per Wing	3 Panels per Wing	4 Panels per Wing
Vmp of 45V BOL	541 x 1198	183 W	366 W	547 W	727 W
	667 x 1198	229 W	457 W	683 W	909 W
	800 x 1198	275 W	548 W	819 W	1090 W
	800 x 1479	320 W	638 W	955 W	1270 W
	800 x 1619	365 W	729 W	1091 W	1451 W
	800 x 1760	411 W	821 W	1228 W	1633 W
	800 x 1900	457 W	911 W	1364 W	1814 W
Vmp of 62V BOL	800 x 1198	263 W	525 W	786 W	1045 W
	800 x 1479	329 W	656 W	981 W	1305 W
	800 x 1619	394W	787 W	1177 W	1566 W
	800 x 1900	460W	917 W	1373 W	1825 W

Selecting the 800mm by 1619mm 1 Panel Per Wing

L4 has a constant with little to **no eclipse time** so batteries are not needed for nominal operations

Battery backup time of **1 hour** is assumed for:

- Irregular operations
- Improving system robustness

Battery capacity requirement of a Depth of Discharge (DOD) of 50% is **944 W*hr.**

The battery selected is the **MP174565 xtd**

Battery power output is **14.6 W*hr**

Equating to a battery pack size of **65 batteries** at 97g totaling **6.3kg for the battery pack**

	ENERGY				EXTENDED LIFE & TEMPERATURE	
	VL 34570 xlr	MP 144350 xlr	MP 174865 xlr	MP 176065 xlr	MP 174565 xtd	MP 176065 xtd
Form factor	Cylindrical D	Prismatic	Prismatic	Prismatic	Prismatic	Prismatic
Nominal voltage	3.65 V	3.65 V	3.65 V	3.65 V	3.65 V	3.65 V
Nominal capacity	5.4 Ah	2.6 Ah	5.3 Ah	6.8 Ah	4.0 Ah	5.6 Ah
Max. continuous discharge current	11.0 A	5.0 A	10.0 A	14.0 A	8.0 A	11.0 A
Max. pulse discharge rate	21.0 A	10.0 A	21.0 A	27.0 A	16.0 A	22.0 A
Max. charge current	5.4 A	2.6 A	5.0 A	6.8 A	4.0 A	5.6 A
Cycle life	>3600	1100	950	1800	2700	2700
(Cycled to 70 % of the cells original capacity)	(100 % DoD, C/2-C/2, + 20°C)	(100 % DoD, C-C/2, + 20°C)	(100 % DoD, C-C/2, + 20°C)	(100 % DoD, C-C/2, + 20°C)	(100 % DoD, C-C/2, + 25°C)	(100 % DoD, C-C/2, + 25°C)
Typical weight	130 g	66 g	121 g	150 g	97 g	135 g
Discharge temperature range	- 35 / + 60°C	- 35 / + 60°C	- 35 / + 60°C	- 35 / + 60°C	- 40 / + 85°C	- 40 / + 85°C
Charge temperature range	- 30 / + 60°C	- 30 / + 60°C	- 30 / + 60°C	- 30 / + 60°C	- 30 / + 85°C	- 30 / + 85°C

Typical values relative to cells stored for one year or less at +30°C max; Performances vary according to discharge characteristics (current, duration, frequency), temperature conditions, storage conditions prior to usage and applications acceptable minimum voltage.



Battery pack is built by Saft Total Energies

Propulsion

Propulsion Requirements and Constraints

Propulsion Requirements
The propulsion subsystem shall provide all station keeping impulses required to maintain the satellite near Sun-Earth L4 for the full mission lifetime.
The propulsion subsystem shall provide a minimum station keeping ΔV capability of 30 m/s over 15 years to account for gravity losses.
The propulsion subsystem shall provide additional delta-v capability for solar radiation pressure correction and trajectory dispersions.
The propulsion subsystem shall account for a minimum 2% unusable propellant residual.
The propulsion subsystem shall maintain partial maneuver capability after the failure of a single thruster.
The propulsion subsystem shall be compatible with launch vehicle hazardous-material and payload safety requirements.
Propulsion Constraints
The propulsion subsystem must operate within the satellite volume allocation for tanks, thrusters, lines, and pressurization hardware.
The propulsion subsystem must not exceed the satellite power budget during thruster valve operation or thermal-control operation.
The propulsion subsystem must be compatible with long duration storage over a 15 year mission lifetime.
The propulsion subsystem must use commercially available or heritage components where possible to reduce development risk and cost.
The propulsion subsystem must minimize total wet mass while still satisfying delta-v, impulse, redundancy, implementation, and lifetime requirements.
The propulsion subsystem must be designed so that thruster plumes minimize contamination of optical payloads, including the coronagraph, heliospheric imager, or other sun observing instruments.

ΔV Calculation

$$\Delta V = I_{sp} g_0 \ln \left(\frac{m_0}{m_f} \right)$$

ΔV = Total Mission Velocity Change

I_{sp} = Thruster Dependent

$g_0 = 9.80665 \frac{m}{s^2}$ = Standard Gravity

$m_0 = 2864 \text{ kg}$ = Dry Mass

m_f = Satellite Wet Mass

Final $\Delta V = 46.5 \frac{m}{s}$

```
% Solve total loaded mass self-consistently
```

```
m_total = m_dry;
```

```
tol = 1e-9;
```

```
for iter = 1:100
```

```
    sol_rad_accel = force_rad / m_total;
```

```
    dv_solar = sol_rad_accel * years_sec;
```

```
    dv_total = dv_station + dv_solar;
```

```
    m_prop_usable = m_dry * (exp(dv_total / ve) - 1);
```

```
    m_prop_margin = m_prop_usable * (1 + margin);
```

```
    m_prop_loaded = m_prop_margin / (1 - ullage);
```

```
    m_total_new = m_dry + m_prop_loaded;
```

```
    if abs(m_total_new - m_total) < tol
```

```
        m_total = m_total_new;
```

```
        break;
```

```
    end
```

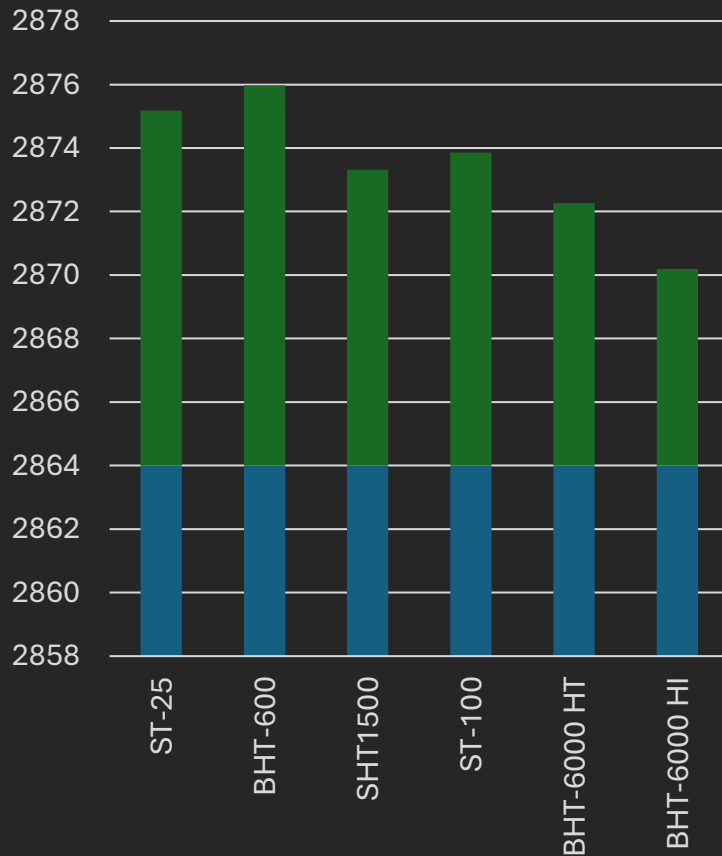
```
    m_total = m_total_new;
```

```
end
```

Satellite and Propellant Mass (kg)

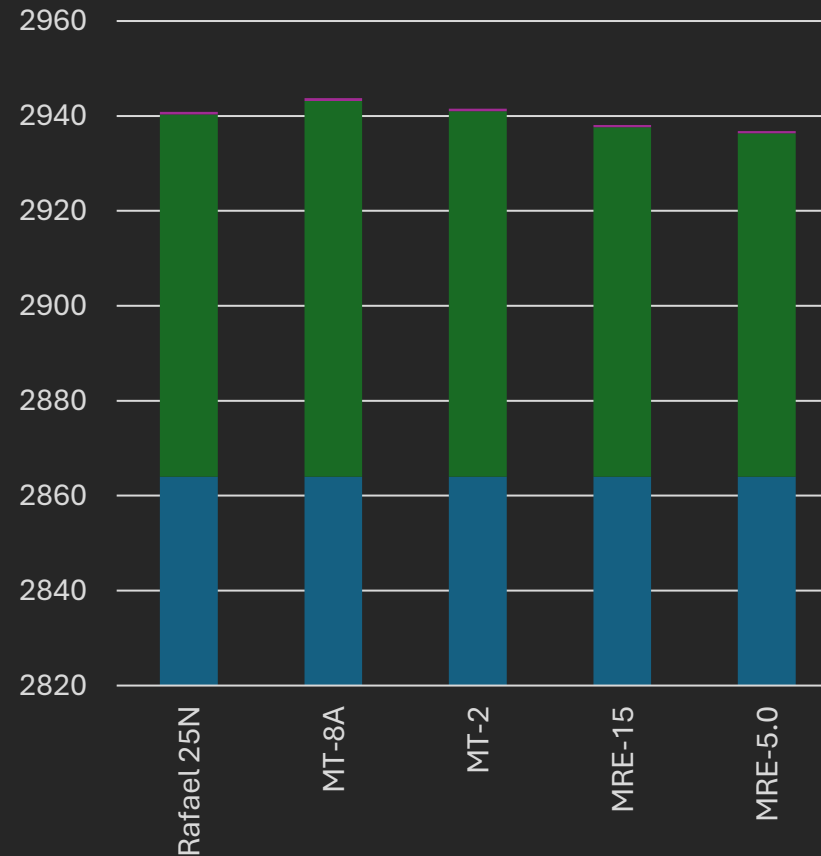
Electric Hall Effect

■ Dry Mass ■ Xenon



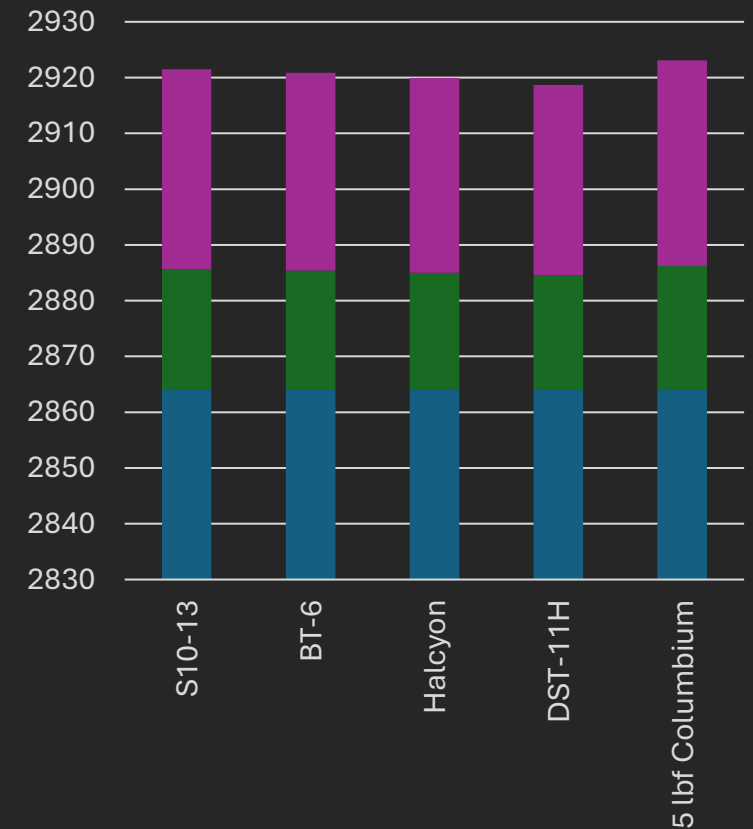
Monopropellant

■ Dry Mass ■ Hydrazine ■ Helium

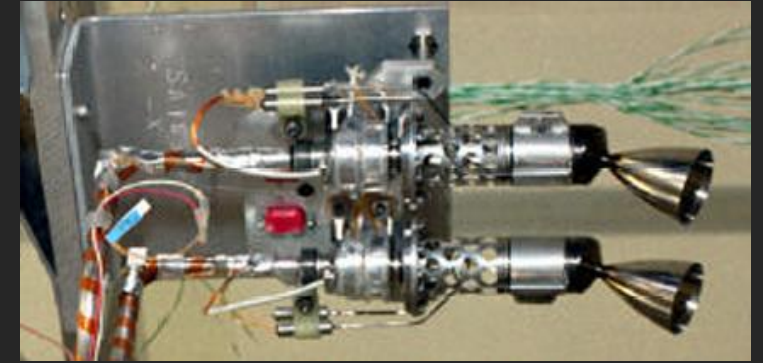


Bipropellant

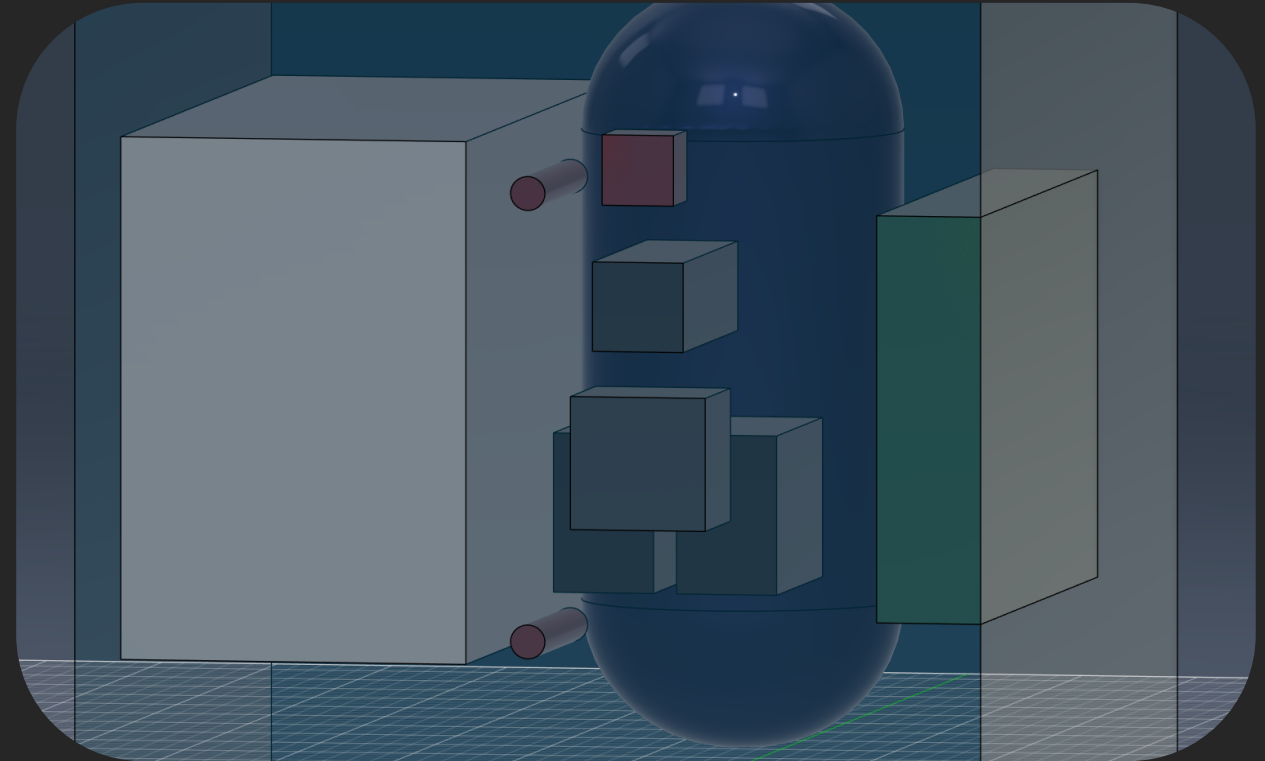
■ Dry Mass ■ MMH ■ MON3



Northrop Grumman MRE-5.0



Propellant	Hydrazine
Thrust	22 N
I_{sp}	232 seconds
Operating Pressure	4.8-27.5 Bar
Life (Maximum Throughput)	456 kg
Power Requirements	19 Watts at 28 Volts
Weight	0.8 kg
Envelope	250mm x 38mm

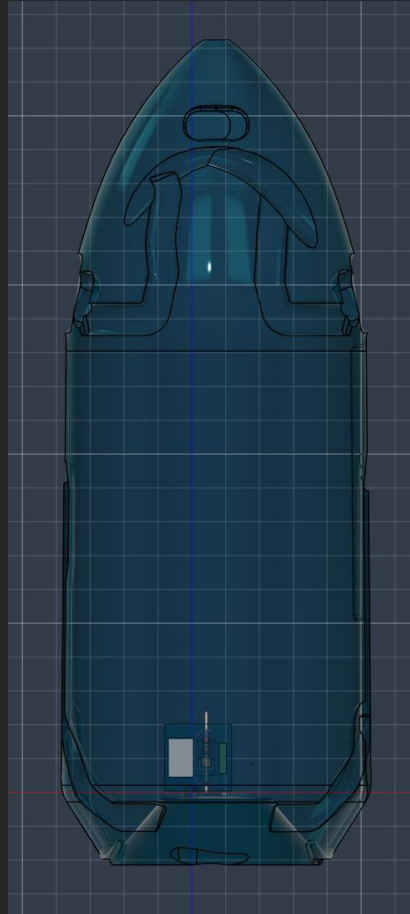


Launch Vehicle

Launch Vehicle Requirements and Constraints

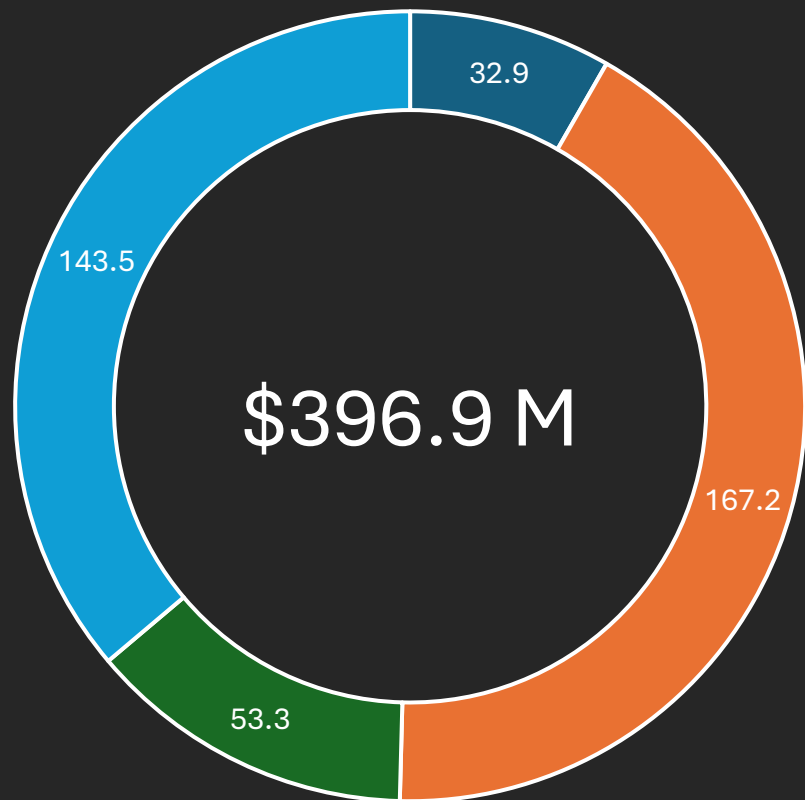
Section	Launch Vehicle Requirements
4.2	The satellite's payload shall comply with the mass and CG properties given.
5.1	The satellite shall be qualified to withstand the shock response induced during payload separation events.
7.2	The satellite primary structure shall be designed to hold a structural lateral frequency above 10 Hz, and a primary axial frequency above 25 Hz, and all secondary structure with a minimum resonant frequency above 35 Hz.
7.2	The satellite shall pass tests such as sine vibration and sine burst tests to ensure structural integrity.
7.2	The satellite structure should be able to handle the maximum steady-state axial and lateral accelerations experienced during flight. The main structure should be heavily reinforced to ensure that the satellite does not collapse.
7.2	During flight, the payload will be subjected to a varying acoustic environment, our satellite shall demonstrate that the payload is not susceptible to acoustic excitation.
Launch Vehicle Constraints	
4.1.3	The payload dimensions must fall within the limits of the static envelope (5.2m x 13.2m for standard fairing or 5.2m x 18.7m for extended fairing).
6.2.1	All cable ties intended for flight must be non-removable, preferably made of Nylon 6/6 or ETFE/Tefzel, and shall be included in vibration test.
6.5	satellite propellant tanks must be designed and evaluated to ensure the safety of the mission. The design shall follow strict safety factors over their Maximum Expected Operating Pressure (MEOP).

Falcon 9



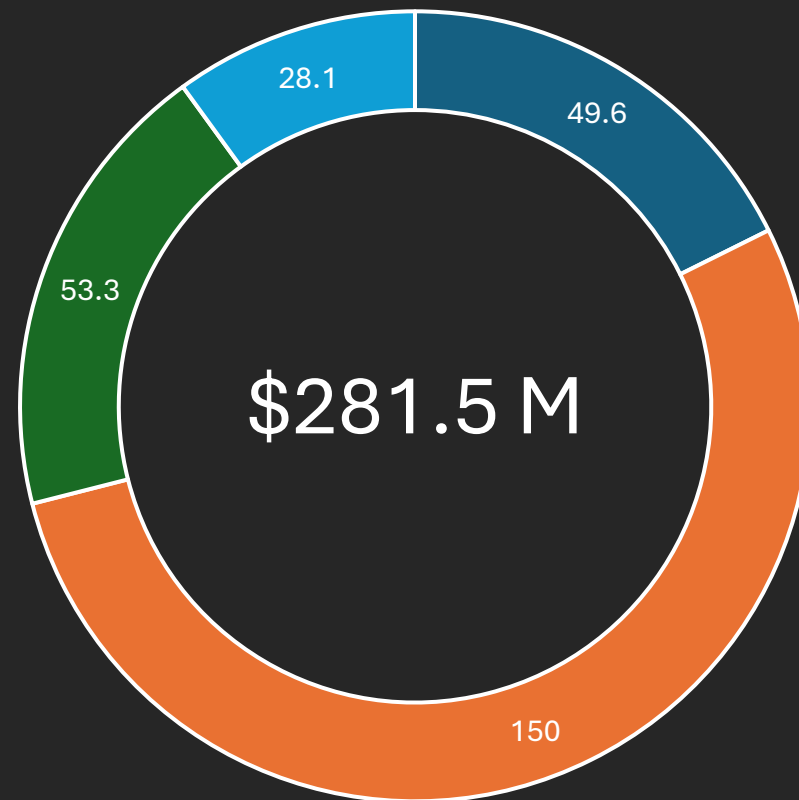
Budget

Full Scientific Payload



■ Flight Sys. And S/C ■ Mission Ops ■ Launch Vehicle ■ Misc.

Minimum Scientific Payload



■ Flight Sys. And S/C ■ Mission Ops ■ Launch Vehicle ■ Misc.

Compliance Matrix

Moon To Mars Strategy and Objectives Development				
Des.	Objectives	Achieved? (Y/N)	Solution	Doc. Section
MI-2	Develop Mars surface, orbital, and Mars-to-Earth communications to support an initial human Mars exploration campaign.	Y		
MI-3	Develop Mars position, navigation and timing capabilities to support an initial human Mars exploration campaign.	N		



OP-10	Demonstrate the capability to operate robotic systems that are used to support crew members on the lunar or Martian surface, autonomously or remotely from the Earth or from orbiting platforms.	N		
AS-1	Characterize and monitor the contemporary environments of the lunar and Martian surfaces and orbits, including investigations of micrometeorite flux, atmospheric weather, space weather, space weathering, and dust, to plan, support, and monitor safety of crewed operations in these locations.	N		
SE-2	Enable Earth-based scientists to remotely support astronaut surface and deep space activities using advanced techniques and tools.	Y		

*See Full Table
Slide (79)*

Decadal Survey Priority Science Goals	% Addressed
How does the Sun maintain its magnetic activity globally from <i>pole to pole</i> ?	100%
How is our <i>home in the galaxy</i> sustained by the Sun and its interaction with the local interstellar medium?	100%

Summary

HEIMDALL will be operating in the Sun-Earth L4 Lagrange point and will minimize communication blackouts between Earth and Mars while also quantifying radiation, plasma, and magnetic field hazards from the Sun.

Subsystem	Mass (Kg)	Daily Power (W)	Additional Resources	Power Margin (20%)
Payload	111.250	136.1		Total Daily Power $\approx$ 471.828
Structures	TBD	0		
Thermal	Min. .650	2		
ADCS	45.790	56.2		
C&DH	5.800	40		
Communications	83.170	149		
Power	19.820	0		
Propulsion	72.80	0	Fuel Tanks (Hydrazine)	
Total	$\approx$ 339.28 +Thermal	$\approx$ 393.21		

Questions?

Additional Slides

Moon To Mars Strategy and Objectives Development				
Des.	Objectives	Achieved? (Y/N)	Solution	Doc. Section
MI-2	Develop Mars surface, orbital, and Mars-to-Earth communications to support an initial human Mars exploration campaign.	Y		
MI-3	Develop Mars position, navigation and timing capabilities to support an initial human Mars exploration campaign.	Y		
LI-2	Develop a lunar surface, orbital, and Moon-to-Earth communications architecture capable of scaling to support long term science, exploration, and industrial needs.	Y		
LI-3	Develop a lunar position, navigation and timing architecture capable of scaling to support long term science, exploration, and industrial needs.	Y		
TH-8	Develop systems that monitor and maintain crew health and performance throughout all mission phases, including during communication delays to Earth, and in an environment that does not allow emergency evacuation or terrestrial medical assistance.	Y		
OP-2	Optimize operations, training and interaction between the team on Earth, crew members on orbit, and a Martian surface team considering communication delays, autonomy level, and time required for an early return to the Earth.	Y		
OP-4	Establish command and control processes, common interfaces, and ground systems that will support expanding human missions at the Moon and Mars.	Y		
OP-10	Demonstrate the capability to operate robotic systems that are used to support crew members on the lunar or Martian surface, autonomously or remotely from the Earth or from orbiting platforms.	N		
AS-1	Characterize and monitor the contemporary environments of the lunar and Martian surfaces and orbits, including investigations of micrometeorite flux, atmospheric weather, space weather, space weathering, and dust, to plan, support, and monitor safety of crewed operations in these locations.	N		
SE-2	Enable Earth-based scientists to remotely support astronaut surface and deep space activities using advanced techniques and tools.	Y		

Mission Schedule

Phase	Start Date (DD/MM/YYYY)	End Date (DD/MM/YYYY)
Development Phase	1/1/2027	1/9/2034
Preliminary Testing	1/9/2034	25/09/2035
Launch / Parking Orbit	25/09/2035	2/10/2035
Phasing Maneuver / L4 Arrival	2/10/2035	1/8/2037
Operational Time	2/8/2037	31/12/2049

Complete Power Consumption Table

Subsystem	Daily Average Power Output (W)	Component (Type)
Payload	136.01	WAVES Heritage (Radio Burst Monitor)
		PLA on VIGIL (Solar Wind Ion/Plasma Analyzer)
		VIGIL Heritage X-ray Monitor (X-ray Flux Monitor)
		MAG/FGM Pair (Fluxgate magnetometer)
		PMI VIGIL (EUV imager/magnetograph)
		CCOR-1 (White-light Coronagraph)
		SOLOHI (Heliospheric Imager)
		STIS (Energy Particle Telescope/Spectrometer)
ADCS	56.2	ASTRO CL (Star Tracker)
		LR-450 IMU (Gyroscope / IMU)
		RDR 68 (Reaction Wheel)
		MONARC-5 (Attitude Control)
		Coarse Sun Sensor (Sun Sensor)
C&DH	40	IPC-7000 (CPU)
Power	10	STARRAY T-STAR (Deployable Solar Array)
		STARBUCK-MINI (SmallSat PCDU / EPS)
Propulsion	0	MRE-5.0 (Monopropellant)
Structure	0	DCUBED HDRM (Solar Panel Deployment)
Thermal	2	Minco Polyimide Thermofoil Kapton Film Heater
Communications	149	TESAT IDT (Transponder)
		MRO TWTA (Amplifier)
		Laser transmitter
		Laser system
Power margin	78.642	20% Power Margin
Total Power	471.852	

Component (Type)	Component Average Daily Power Output	Theoretical Max Power Output	Station Keeping	Deployment
WAVES Heritage (Radio Burst Monitor)	8	8	0	0
PLA on VIGIL (Solar Wind Ion/Plasma Analyzer)	14	14	0	0
VIGIL Heritage X-ray Monitor (X-ray Flux Monitor)	30	30	0	0
MAG/FGM Pair (Fluxgate magnetometer)	1.71	1.71	0	0
PMI VIGIL (EUV imager/magnetograph)	40	40	0	0
CCOR-1 (White-light Coronagraph)	25	25	0	0
SOLOHI (Heliospheric Imager)	13.5	13.5	0	0
STIS (Energy Particle Telescope/Spectrometer)	3.8	3.8	0	0
ASTRO CL (Star Tracker)	1.2	2.4	1.2	0
LR-450 IMU (Gyroscope / IMU)	15	30	15	15
RDR 68 (Reaction Wheel)	40	360	40	40
MONARC-5 (Attitude Control)	0	144	144	6
Coarse Sun Sensor (Sun Sensor)	0	6	6	6
IPC-7000 (CPU)	40	80	40	40
STARRAY T-STAR (Deployable Solar Array)	0	0	0	0
STARBUCK-MINI (SmallSat PCDU / EPS)	10	10	10	10
MRE-5.0 (Monopropellant)	0	76	76	0
DCUBED HDRM (Solar Panel Deployment)	0	128	0	128
Minco Polyimide Thermofoil Kapton Film Heater	2	10	2	2
TESAT IDT (Transponder)	20	20	20	20
MRO TWTA (Amplifier)	50	100	0	0
Laser transmitter	4	4	0	0
Laser system	75	100	0	0
20% Power Margin	78.642	181.74	70.84	53.4
Power (W)	471.852	1388.15	425.04	320.4

ADCS SWaP we can delete this later/move it to the SWaP

Component	Quantity	Unit Dimensions	Unit Mass	Total Mass	Unit Power
RDR-68 Reaction Wheel	4	345 mm dia × 118 mm h	7.6 kg	30.4 kg	<20 W steady
RDR-68 Wheel Drive Electronics	4	52 × 247 mm ground plane, 145 mm h	1.25 kg	5.0 kg	included with wheel power
LR-450 IMU	1	8.6 × 6.3 × 6.3 in	10 lb (4.54 kg)	4.54 kg	<15 W
ASTRO CL Star Tracker	2	60 × 60 × 129 mm	0.32 kg	0.64 kg	<1.2 W
Coarse Sun Sensor	6	110 × 110 × 30 mm	0.215 kg	1.29 kg	0 W
MONARC-5 Thruster	8	41.8 cm long, 2.5 cm nozzle exit dia	0.49 kg	3.92 kg	18 W

SWAP

Science & Communication Goals	Equipment Type	Proposed Instrument	Instrument Dimensions (m^3,g)				Max Power req.	
			LxWxH (m)	Volume (m^3)	Mass (g)	%	(W)	%
Enable Early Space Weather Detection	Coronagraph	Compact Coronagraph (CCOR-1)	0.889	0.197873798	25000	15%	22.5	3.92%
			0.5842					
			0.381					
	Energetic Particle Telescope	STIS		0	2300		3.8	
	Radio Burst Monitor	WAVES Heritage	50	0.125	8000		8	
			0.05					
			0.05					
	Solar Wind Ion/Plasma Analyzer	PLA on VIGIL		0.07875	16500		14	
	Heliospheric Imager	SOLOHI Heritage	0.684	1.07875	16600		13.5	
			0.455					
			0.115					
	X-ray Flux Monitor	VIGIL X-ray Monitor	0.1	2.07875	14000		30	
			0.1					
			0.32					
	EUV Imager/Magnetograph	PMI VIGIL		0.1487	28200		40	
	Fluxgate Magnetometer	MAG/FGM Pair	0.02	1.1487	650		1.71	
			0.02					
			0.02					
Establish and Maintain Consistent Comms between earth and Mars.	TWTA (Traveling Wave Tube Amp)	Heritage TWTA (using MRO TWTA for reference)	0.4826	0.092627879	9500	5.67%	100	17.42%
			0.22225					
			0.8636					
	Transponder	TESAT IDT or SDST alternative	0.22	0.0033	2000	1.19%	20	3.48%
			0.1					
			0.15					
	HGA(High Gain Antenna)	MRO HGA reference	2	4	19100	11.39%	0	0.00%
			2					
			1					
	Data Processing	Sirius Quadcore LEON4FT	0.1	0.00015	100	0.06%	4	0.70%
			0.1					
			0.015					
	Data Processing	IADCS200	0.95	0.2736	470	0.28%	4.5	0.78%
			0.9					
			0.32					
	Transmitter	TESAT IDT	0.22	0.0033	2000	1.19%	20	3.48%
			0.1					
			0.15					
	Transmitter/receiver	TESAT SCOT135	0.203	0.0123	50000	86.86%	75	59.23%
			0.225					
			0.27					

	Equipment Type	Proposed Instrument	Instrument Dimensions (m^3,g)				Max Power req.	
			LxWxH (m)	Volume (m^3)	Mass (g)	%	(W)	%
Subsystems	ADCS	RDR-68 Reaction Wheel	0.345 0.345 0.118 0.052	0.01404495	30400		20	
		RDR-68 Wheel Drive Electronics	0.247 0.145 0.218 0.16	0.00186238	5000		-	
		LR-450 IMU	0.16 0.16 0.06 0.06 0.129 0.11	0.0055808	4540		15	
		ASTRO CL Star Tracker	0.06 0.06 0.129 0.11 0.11 0.03	0.0004644	640		1.20	
		Coarse Sun Sensor	0.11 0.03 0.418 0.025 0.025 0.25	0.000363	1290		0	
		MONARC-5 Thruster	0.038 0.038 0.8 1.619 1	0.00026125	3920		144	
		MRE-5.0	0.038 0.038 0.8 1.619 1	0.000361	3200		76	
		T-Star Solar Array	0.038 0.038 0.8 1.619 1	0.000361	3200		76	
		Saft Total Energies MP 176065 xtd	0.038 0.038 0.8 1.619 1	0.000361	3200		76	
		Multi-Layer Insulation (MLI) Blankets	0.038 0.038 0.8 1.619 1	0.000361	3200		76	
	Propulsion	MRE-5.0	0.038 0.038 0.8 1.619 1	0.000361	3200		76	
		T-Star Solar Array	0.038 0.038 0.8 1.619 1	0.000361	3200		76	
		Saft Total Energies MP 176065 xtd	0.038 0.038 0.8 1.619 1	0.000361	3200		76	
		Multi-Layer Insulation (MLI) Blankets	0.038 0.038 0.8 1.619 1	0.000361	3200		76	
		AZ-93 Zinc Oxide-based White Coating	0.038 0.038 0.8 1.619 1	0.000361	3200		76	
	Power	Aluminum Heat Pipes	0.038 0.038 0.8 1.619 1	0.000361	3200		76	
		AZ-93 High-Emissivity Coated Aluminum Radiator Panels	0.038 0.038 0.8 1.619 1	0.000361	3200		76	
		Minco Polyimide Thermofoil Kapton Film Heaters	0.038 0.038 0.8 1.619 1	0.000361	3200		76	
		Voyager Technologies IPC-7000	0.038 0.038 0.8 1.619 1	0.000361	3200		76	
		Satellite Main Body	0.038 0.038 0.8 1.619 1	0.000361	3200		76	
	Thermal	Aluminum Heat Pipes	0.038 0.038 0.8 1.619 1	0.000361	3200		76	
		AZ-93 High-Emissivity Coated Aluminum Radiator Panels	0.038 0.038 0.8 1.619 1	0.000361	3200		76	
		Minco Polyimide Thermofoil Kapton Film Heaters	0.038 0.038 0.8 1.619 1	0.000361	3200		76	
		Voyager Technologies IPC-7000	0.038 0.038 0.8 1.619 1	0.000361	3200		76	
		Satellite Main Body	0.038 0.038 0.8 1.619 1	0.000361	3200		76	
	CD&H	Heritage TWTA (using MRO TWTA for reference)	0.268 0.178 0.01105 1.15 1.15 1.15 0.4826 0.22225 0.8636 0.22 0.1 0.15 2 2 1 0.1 0.1 0.015 0.95 0.9 0.32 0.22 0.1 0.15 0.203 0.225 0.27	0.001054258	5800		40	
		Satellite Main Body	0.268 0.178 0.01105 1.15 1.15 1.15 0.4826 0.22225 0.8636 0.22 0.1 0.15 2 2 1 0.1 0.1 0.015 0.95 0.9 0.32 0.22 0.1 0.15 0.203 0.225 0.27	0.001054258	5800		40	
		Heritage TWTA (using MRO TWTA for reference)	0.268 0.178 0.01105 1.15 1.15 1.15 0.4826 0.22225 0.8636 0.22 0.1 0.15 2 2 1 0.1 0.1 0.015 0.95 0.9 0.32 0.22 0.1 0.15 0.203 0.225 0.27	0.001054258	5800		40	
		TESAT IDT or SDST alternative	0.268 0.178 0.01105 1.15 1.15 1.15 0.4826 0.22225 0.8636 0.22 0.1 0.15 2 2 1 0.1 0.1 0.015 0.95 0.9 0.32 0.22 0.1 0.15 0.203 0.225 0.27	0.001054258	5800		40	
		MRO HGA reference	0.268 0.178 0.01105 1.15 1.15 1.15 0.4826 0.22225 0.8636 0.22 0.1 0.15 2 2 1 0.1 0.1 0.015 0.95 0.9 0.32 0.22 0.1 0.15 0.203 0.225 0.27	0.001054258	5800		40	
	Structures	Satellite Main Body	0.268 0.178 0.01105 1.15 1.15 1.15 0.4826 0.22225 0.8636 0.22 0.1 0.15 2 2 1 0.1 0.1 0.015 0.95 0.9 0.32 0.22 0.1 0.15 0.203 0.225 0.27	0.001054258	5800		40	
		Heritage TWTA (using MRO TWTA for reference)	0.268 0.178 0.01105 1.15 1.15 1.15 0.4826 0.22225 0.8636 0.22 0.1 0.15 2 2 1 0.1 0.1 0.015 0.95 0.9 0.32 0.22 0.1 0.15 0.203 0.225 0.27	0.001054258	5800		40	
		TESAT IDT or SDST alternative	0.268 0.178 0.01105 1.15 1.15 1.15 0.4826 0.22225 0.8636 0.22 0.1 0.15 2 2 1 0.1 0.1 0.015 0.95 0.9 0.32 0.22 0.1 0.15 0.203 0.225 0.27	0.001054258	5800		40	
		MRO HGA reference	0.268 0.178 0.01105 1.15 1.15 1.15 0.4826 0.22225 0.8636 0.22 0.1 0.15 2 2 1 0.1 0.1 0.015 0.95 0.9 0.32 0.22 0.1 0.15 0.203 0.225 0.27	0.001054258	5800		40	
		Satellite Main Body	0.268 0.178 0.01105 1.15 1.15 1.15 0.4826 0.22225 0.8636 0.22 0.1 0.15 2 2 1 0.1 0.1 0.015 0.95 0.9 0.32 0.22 0.1 0.15 0.203 0.225 0.27	0.001054258	5800		40	
	Communications	Heritage TWTA (using MRO TWTA for reference)	0.268 0.178 0.01105 1.15 1.15 1.15 0.4826 0.22225 0.8636 0.22 0.1 0.15 2 2 1 0.1 0.1 0.015 0.95 0.9 0.32 0.22 0.1 0.15 0.203 0.225 0.27	0.001054258	5800		40	
		TESAT IDT or SDST alternative	0.268 0.178 0.01105 1.15 1.15 1.15 0.4826 0.22225 0.8636 0.22 0.1 0.15 2 2 1 0.1 0.1 0.015 0.95 0.9 0.32 0.22 0.1 0.15 0.203 0.225 0.27	0.001054258	5800		40	
		MRO HGA reference	0.268 0.178 0.01105 1.15 1.15 1.15 0.4826 0.22225 0.8636 0.22 0.1 0.15 2 2 1 0.1 0.1 0.015 0.95 0.9 0.32 0.22 0.1 0.15 0.203 0.225 0.27	0.001054258	5800		40	
		Satellite Main Body	0.268 0.178 0.01105 1.15 1.15 1.15 0.4826 0.22225 0.8636 0.22 0.1 0.15 2 2 1 0.1 0.1 0.015 0.95 0.9 0.32 0.22 0.1 0.15 0.203 0.225 0.27	0.001054258	5800		40	
		Heritage TWTA (using MRO TWTA for reference)	0.268 0.178 0.01105 1.15 1.15 1.15 0.4826 0.22225 0.8636 0.22 0.1 0.15 2 2 1 0.1 0.1 0.015 0.95 0.9 0.32 0.22 0.1 0.15 0.203 0.225 0.27	0.001054258	5800		40	
		TESAT IDT or SDST alternative	0.268 0.178 0.01105 1.15 1.15 1.15 0.4826 0.22225 0.8636 0.22 0.1 0.15 2 2 1 0.1 0.1 0.015 0.95 0.9 0.32 0.22 0.1 0.15 0.203 0.225 0.27	0.001054258	5800		40	
		MRO HGA reference	0.268 0.178 0.01105 1.15 1.15 1.15 0.4826 0.22225 0.8636 0.22 0.1 0.15 2 2 1 0.1 0.1 0.015 0.95 0.9 0.32 0.22 0.1 0.15 0.203 0.225 0.27	0.001054258	5800		40	
		Satellite Main Body	0.268 0.178 0.01105 1.15 1.15 1.15 0.4826 0.22225 0.8636 0.22 0.1 0.15 2 2 1 0.1 0.1 0.015 0.95 0.9 0.32 0.22 0.1 0.15 0.203 0.225 0.27	0.001054258	5800		40	
		Heritage TWTA (using MRO TWTA for reference)	0.268 0.178 0.01105 1.15 1.15 1.15 0.4826 0.22225 0.8636 0.22 0.1 0.15 2 2 1 0.1 0.1 0.015 0.95 0.9 0.32 0.22 0.1 0.15 0.203 0.225 0.27	0.001054258	5800		40	
		TESAT IDT or SDST alternative	0.268 0.178 0.01105 1.15 1.15 1.15 0.4826 0.22225 0.8636 0.22 0.1 0.15 2 2 1 0.1 0.1 0.015 0.95 0.9 0.32 0.22 0.1 0.15 0.203 0.225 0.27	0.001054258	5800		40	
		MRO HGA reference	0.268 0.178 0.01105 1.15 1.15 1.15 0.4826 0.22225 0.8636 0.22 0.1 0.15 2 2 1 0.1 0.1 0.015 0.95 0.9 0.32 0.22 0.1 0.15 0.203 0.225 0.27	0.001054258	5800		40	

Compliance Matrix

Moon To Mars Strategy and Objectives Development				
Des.	Objectives	Achieved? (Y/N)	Solution	Doc. Section
MI-2	Develop Mars surface, orbital, and Mars-to-Earth communications to support an initial human Mars exploration campaign.	Y		
MI-3	Develop Mars position, navigation and timing capabilities to support an initial human Mars exploration campaign.	N		
LI-2	Develop a lunar surface, orbital, and Moon-to-Earth communications architecture capable of scaling to support long term science, exploration, and industrial needs.	Y		
LI-3	Develop a lunar position, navigation and timing architecture capable of scaling to support long term science, exploration, and industrial needs.	Y		
TH-8	Develop systems that monitor and maintain crew health and performance throughout all mission phases, including during communication delays to Earth, and in an environment that does not allow emergency evacuation or terrestrial medical assistance.	N		
OP-2	Optimize operations, training and interaction between the team on Earth, crew members on orbit, and a Martian surface team considering communication delays, autonomy level, and time required for an early return to the Earth.	Y		
OP-4	Establish command and control processes, common interfaces, and ground systems that will support expanding human missions at the Moon and Mars.	Y		
OP-10	Demonstrate the capability to operate robotic systems that are used to support crew members on the lunar or Martian surface, autonomously or remotely from the Earth or from orbiting platforms.	N		
AS-1	Characterize and monitor the contemporary environments of the lunar and Martian surfaces and orbits, including investigations of micrometeorite flux, atmospheric weather, space weather, space weathering, and dust, to plan, support, and monitor safety of crewed operations in these locations.	N		
SE-2	Enable Earth-based scientists to remotely support astronaut surface and deep space activities using advanced techniques and tools.	Y		

Objectives	Requirements	Capabilities	Comply? (Y/N/Deg)	Doc. Section
PSG1: How does the Sun maintain its magnetic activity globally from pole to pole?				
1.a. Determine the role of the polar and high latitude magnetic fields and flows in the evolution of the solar dynamo.	Vector observations of magnetic fields and flows in polar regions for at least 1 month on each pole at a 5-minute cadence (1.a, 1.b, 1.c)	Long-term (several months) spectropolarimetric observations of the polar regions. High resolution Doppler measurements.	Y	
	Magnetic flux emergence processes and connection to global corona (1.a, 1.b, 1.c)	Global synchronous (front and back side) multiviewpoint magnetogram and doppler measurements	Y	
	Helioseismic measurements of tachocline thickness, global flows, inertial waves at all latitudes and depths (1.a, 1.b, 1.c)	Multiyear multiviewpoint doppler measurements including equatorial nearside, far-side, and polar regions	Y	
	Joint surface and helioseismic measurements. Modernize historical data to produce a long-term homogeneous observational baseline (1.a, 1.b, 1.c, 1.d)	Modernization, intercalibration, and homogenization of historical records, providing a multidecadal, multicycle observational set	Y	
1.d. Determine how solar magnetism changes over the solar cycle, from the solar interior, and through the highly coupled solar atmosphere.				
PSG2: How do the Sun's magnetic fields and radiation environments connect throughout the heliosphere?				
2.a. Demonstrate how photospheric and chromospheric dynamics drive the corona.	Magnetic field strengths and topology in the solar atmosphere.	High spatial resolution, high cadence optical and NIR imaging spectropolarimetry	Y	
		Combination of radio, O/IR, EUV spectropolarimetry	Y	
2.b. Determine the role quasi-steady processes play in the heating of the solar corona and the acceleration of the solar wind and nonthermal particles.	Photospheric drivers: power spectra of MHD waves, and tangling of field lines	High spatial resolution, high cadence optical imaging spectroscopy	N	
	Thermal response of upper solar atmosphere	Remote: high resolution, high cadence radio, optical/IR, EUV, and SXR imaging spectroscopy In situ: multipoint, multiscale velocity vector, density, temperature measurements	Y	
2.c. Understand how the magnetic field of the corona and inner heliosphere is structured, how it evolves, and how it connects to and influences the interplanetary magnetic field on varying timescales.	Spatial, temporal, and spectral distribution of energetic particles and their composition.	Remote: high-dynamic-range radio and HXR imaging spectropolarimetry	N	
		In situ: multipoint, multiscale electron spectral measurements	Y	
		Remote: sensitive gamma ray imaging spectroscopy	N	
		In situ: multipoint, multiscale, multispecies electron/ion/neutron spectral measurements and composition	Deg	
2.d. Trace the origin of solar wind variability, and identify the extent to which it is owing to local kinetic processes or underlying global solar activity.	Time varying and spatially distributed measurements of the heliospheric magnetic field. Distribution, dynamics, composition, and thermal properties of magnetized plasma (protons, alphas and heavy ions), suprathermal properties.	Remote: EUV measurements, magnetograms, spectroscopy, coronal fields, heliospheric imaging	Y	
		In situ: multipoint, multiscale, multispecies electron/ion/neutron energy measurements, velocity distributions, composition, and in situ magnetic fields, radio waves.	Y	

PSG3: How do solar explosions unleash their energy throughout the heliosphere?				
3.a. Understand the dominant energy conversion and transport mechanisms that energize plasma and particles throughout the heliosphere. 3.b. Measure and track the rapidly evolving properties of solar eruptions from the solar surface through interplanetary space. 3.c. Measure and track the rapidly evolving properties of solar eruptions from the solar surface through interplanetary space. 3.d. Improve methods for forecasting and nowcasting of solar eruptions at the Sun and for predicting the subsequent impacts on interplanetary radiation environments	Vector magnetic field in the solar atmosphere before, during, and after the events.	High spatial resolution, high cadence optical imaging spectropolarimetry.	Y	
		High spatial resolution, high cadence infrared imaging spectropolarimetry	Y	
		Combination of radio, O/IR, EUV spectropolarimetry	Y	
	Spatial, temporal, and spectral distribution of energetic particles and their composition.	Remote: high-dynamic-range radio and HXR imaging spectropolarimetry	N	
		In situ: multipoint, multiscale electron spectral measurements	Y	
		Remote: sensitive gamma ray imaging spectroscopy	N	
	Distribution, dynamics, composition, and thermal properties of magnetized plasma.	In situ: multipoint, multiscale, multispecies ion/neutron spectral measurements and composition	Deg	
		Remote: high resolution, high cadence radio, optical/IR, EUV, and SXR imaging spectroscopy	Y	
		In situ: multipoint, multiscale velocity vector, density, temperature measurements for electrons, ions, including heavy ions	Deg	
	Initiation, distribution, and evolution of shocks and waves.	Remote: dm-m radio imaging spectroscopy, multi-LOS Faraday rotation observations	N	
		Remote: White-light coronagraph imaging from multiple perspectives	Y	
		In situ: multipoint, high-cadence measurements of B, n, T, v	Deg	
PSG4: How is our home in the galaxy sustained by the Sun and its interaction with the local interstellar medium?				
4.a. Ascertain the physical processes from the Sun to the LISM that shape the heliosphere and determine the spatial and temporal dependance of boundaries 4.b. Establish how the dynamics and evolution of the global heliosphere are affected by solar activity, and by the LISM and its inhomogeneities. 4.c. Determine how pickup ions are heated and accelerated across the termination shock, evolve in the heliosheath, and escape across the heliopause into the LISM. 4.d. Infer how and where anomalous cosmic rays are accelerated, and how they and galactic cosmic rays are modulated by the Sun, heliosphere, and LISM.	Spatial structure and temporal variations of the heliosheath and LISM through remote sensing from inside and outside the heliosphere.	ENA imaging of heliosheath and ribbon from a changing vantage point	Deg	
		Unraveling of processing and filtration of interstellar hydrogen, helium and other species by the heliosphere; properties of pristine interstellar neutrals	Deg	
		filtration of interstellar hydrogen, helium and other species by the heliosphere; properties of pristine interstellar neutrals	Deg	
		Interstellar dust composition along outward trajectory from 1 AU to LISM	Y	
	The physical processes that form the Ribbon and determine its extent, shape, and temporal variability.	ENA imaging from a changing vantage point	Deg	
		Ion, neutral and magnetic field observations through the ribbon	Y	
	The complete ion distribution function at thermal, suprathermal and high energies, across the termination shock and throughout the heliosheath, and across the heliopause where they escape.	Thermal, pick-up (suprathermal), and energetic ion distribution and composition, vector magnetic fields along outward trajectory from 1 AU to LISM	Y	
	Particle acceleration, energy dissipation, shocks and transients throughout heliosphere and LISM.	Turbulence and waveparticle measurements along outward trajectory from 1 AU to LISM	Y	
	Spatial and temporal variation of the spectra and anisotropy of galactic and anomalous cosmic rays, throughout the heliosphere and in the LISM.	GCR and ACR fluxes, spectra, composition and anisotropy into the LISM	Y	90

Mission Schedule

Phase	27	28	29	30	31	32	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	49	
Development Phase	2027 - 2034																							
Preliminary Testing									'34-'35															
Launch / Parking Orbit									'35															
Phasing Maneuver / L4 Arrival									2035 - 2037															
Operational Time											2037 - 2049													

Objective

Feasibility

Expense

Novelty

Trojan Asteroid
Op. Location

Venus or Mercury
Lagrange points

Earth L5
Position

Earth L4 Position

